

The AI-GPR Index: Measuring Geopolitical Risk using Artificial Intelligence*

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Abstract

We introduce an improved measure of geopolitical risk, constructed by applying artificial intelligence to newspaper articles. Our approach replaces traditional keyword-based methods with semantic understanding: instead of searching for specific word combinations, we use OpenAI’s GPT-4o mini model to read newspaper articles and assess their geopolitical risk intensity. The daily AI-GPR index scores articles from the New York Times, Washington Post, and Chicago Tribune from 1960 to the present, around 4.6 million of them over 1960–2025. The approach reduces false positives from articles mentioning war or terrorism in non-geopolitical contexts while capturing relevant articles that lack exact or common dictionary terms. The AI-GPR index also assigns gradations of risk intensity rather than simple binary yes-or-no classifications, providing more nuanced measurement. We demonstrate the potential of our approach with several applications. The AI-GPR index produces a more precisely estimated negative effect of geopolitical risk on stock returns, and decomposing it into anticipated threats and realized acts reveals that markets price the two differently. A second LLM classification layer yields a time series of geopolitical risk-driven oil supply disruptions by region, which we use to trace the macroeconomic effects of geopolitical oil shocks. Finally, we decompose geopolitical risk by country and by directed country pair, and show that a rise in bilateral geopolitical risk is followed by a contraction in trade between the two countries.

JEL Classification: D80, E66, F51, G15.

Keywords: Geopolitical Risk, Textual Analysis, Large Language Models, Measurement Error, Content Classification.

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1 Introduction

Geopolitical risk—encompassing wars, terrorism, and tensions between nations—is an important driver of macroeconomic fluctuations. Measuring its effects requires quantifying latent political tensions from sparse and noisy data. [Caldara and Iacoviello \(2022\)](#) established a framework for this measurement with the Geopolitical Risk (GPR) Index, using dictionary-based keyword counts to measure geopolitical instability through newspaper archives.¹ Their index, measured as the share of newspaper articles discussing geopolitical risks, has since become a standard benchmark in the literature as a proxy for the risks that shape trade patterns, investor decisions, and asset prices.

While keyword-based methods have proven effective at capturing global risks and other economic phenomena, they have limits. They struggle to distinguish articles where a geopolitical event is the primary subject from those where keywords appear in only a peripheral or historical context. Fixed dictionaries may also miss evolving terminology or fail to capture the nuance of diplomatic language. This paper keeps the definition of geopolitical risk in [Caldara and Iacoviello \(2022\)](#) but replaces keyword matching with a language model that reads each article.

We introduce the AI-GPR Index, constructed by applying a Large Language Model (LLM), specifically OpenAI’s GPT-4o mini, to articles from the *New York Times*, *Washington Post*, and *Chicago Tribune* from 1960 to the present. Over 1960–2025, the corpus contains approximately 4.6 million articles. Our approach maintains the definition of geopolitical risk established in the prior literature but uses the LLM to score articles on a continuous scale, so that the index distinguishes the severity of geopolitical events as the news presents them. The resulting index complements the original GPR by offering a more accurate measure that retains the transparency of the original framework through auditable prompts and classification criteria. It is also less noisy. Around major events such as the Cuban Missile Crisis, the September 11 attacks, and Russia’s invasion of Ukraine, both indices spike at the onset, but the daily AI-GPR moves much less from one day to the next, because the LLM does not need specific words to appear in an article to recognize geopolitical risk. This makes the AI-GPR better suited to financial applications that rely on high-frequency data.

We demonstrate the value of the AI-GPR index through a number of empirical applications. First, we revisit the relationship between geopolitical risk and financial markets. We find that the AI-GPR index provides a more precisely estimated negative effect of geopolitical innovations on stock market returns, particularly when isolating the persistent component of risk. A further classification layer decomposes the index into anticipated threats and realized acts, and we find that the two are priced differently: markets price the gradual and persistent buildup of threats, and respond to realized acts only insofar as they are unanticipated.

¹ The use of dictionary-based methods itself builds on the foundational work of [Baker, Bloom, and Davis \(2016\)](#) and [Hassan et al. \(2019\)](#), among others.

Second, we apply a second LLM classifier to identify articles about geopolitical oil supply disruptions. This yields a time series of GPR-driven oil shocks that can be disaggregated by geographic region. We show an application of the “Oil GPR index” using a proxy structural VAR, in the spirit of [Verduzco-Bustos and Zanetti \(2026\)](#), but replacing their original GPR instrument with a proxy constructed from oil price surprises on days when threats of geopolitical oil supply disruptions spike; the identified shock generates a persistent rise in oil prices, a rise in inflation, and a decline in economic activity.

Third, we use the LLM to identify the country-level actors involved in geopolitical events and their roles as initiators, respondents, or spillover parties. We use these labels to reconstruct the directed actions between country-level actors. This yields country-level indices that track the risk involving each country over time, extending the country sub-indices of [Caldara and Iacoviello \(2022\)](#) to two hundred nations, along with high-frequency, directed country-pair indices that capture the source and target of bilateral tensions. By mapping these directed links, we can also identify the specific country pairs that drive global risk. We validate the bilateral index by estimating its effects on bilateral trade flows. Trade adjusts with a delay, so we trace the response with a local projection: a rise in bilateral risk is followed by a decline in trade that builds over two years and then fades.

We make all the measures presented in this paper available in a regularly updated companion webpage. Several of them—the country-level series, the oil-market series, and the bilateral series—respond to requests from researchers, policymakers, and market participants who use the original index of [Caldara and Iacoviello](#). The index is live: new articles are scored as they are published, while the empirical exercises in this paper use fixed samples whose end dates we report in [Section 2](#).

Our work contributes to a rapidly emerging literature applying LLMs to economic measurement (e.g., [Ito, Sato, and Ota, 2025](#) and [Clayton, Coppola, Maggiori, and Schreger, 2025](#)). We show that LLM-based approaches augment rather than replace dictionary-based methods. Our focus on geopolitical risk also connects our work to the growing literature on geoeconomics (see [Mohr and Trebesch, 2025](#) and [Clayton, Maggiori, and Schreger, 2026](#)) and provides an improved set of data to make sense of geopolitical risk both within and across country borders.

[Section 2](#) describes the data and the LLM-based methodology. [Section 3](#) presents the AI-GPR index. [Section 4](#) assesses the index’s validity and robustness, contrasting it with keyword classification, validating it against human coding, bounding measurement error, and confirming insensitivity to the choice of model. [Section 5](#) introduces three further dimensions of measurement: threats and acts, oil, and country and bilateral risk. [Section 6](#) presents our empirical applications, and [Section 7](#) concludes.

2 Data and Methodology

2.1 Data Sources

Our construction of the AI-GPR index sources newspaper data from the New York Times, Washington Post, and Chicago Tribune, all of which report on international affairs throughout the sample. The articles we score span from 1960 to the present and are sampled daily.

The AI-GPR is a live index. As new articles are published, we score them with the same query, prompt, and model, and the companion webpage posts the updated series. Any paper, however, has to work with a fixed vintage of the data, and different parts of this one use different samples. The corpus counts, summary statistics, and computational cost refer to articles published between 1960 and 2025, the period covered by the search query below, and so do the human audit and the model comparison of Section 4. The time-series figures and the stock-return regressions of Section 6.1 use the index through March 2026. The oil VAR of Section 6.2 runs from 1985 to 2024, and the trade analysis of Section 6.3 from 1962 to 2020, with samples limited by the other data these applications require.

We adopt a two-stage process that first filters for articles likely to discuss geopolitical events and then passes them through an LLM for scoring. The keyword screen cheaply discards articles with no geopolitical content, and the LLM then scores the remaining articles more accurately than a query could.

We use the following query on the ProQuest TDM Studio database to filter for articles likely to discuss geopolitical risk:

SEARCH QUERY:

```
PUBDATE(>=19600101) AND PUBDATE(<=20251231) AND LA(English) AND (airstrike* OR alliance OR annex* OR attack* OR blockade* OR bomb* OR cease-fire OR combat OR confrontation OR conflict* OR coup OR crisis OR diplomat* OR diplomacy OR embargo OR enem* OR hostage* OR hostile* OR instability OR invade* OR invasion* OR militia OR military OR missile* OR nuclear OR refugee* OR riot* OR sanction* OR sovereign* OR terror* OR treaty OR troop* OR truce OR unrest OR violence OR war OR weapon*) AND (publication("New York Times") OR publication("Washington Post") OR publication("Chicago Tribune")) AND DTYPE(article OR commentary OR editorial OR feature OR 'front page article' OR 'front page/cover story' OR news OR report OR review)
```

The query focuses on articles containing at least one keyword potentially related to geopolitical risk. We further restrict our attention to articles categorized as news content rather than

advertisements or non-editorial material. Our keywords span three categories of geopolitical risk: direct conflict and military action (war, invasion*, attack*, terror*, conflict*, ...), political crisis and instability (crisis, instability, unrest, coup...), and strategic responses and diplomacy (sanction*, embargo, blockade*...). Our search extracts 4.6 million articles from the universe of articles published across the three publications over the sample period. Of these 4.6 million articles, 1.2 million (26 percent) receive a positive GPR score from the LLM. Our search thus retrieves 54 percent of all the articles published in the three newspapers over the sample period.² Section 4.3 verifies that the keyword filter has a negligible false-omission rate.

For each selected article, we extract metadata (publication date, headline) and its textual content from the ProQuest database. To manage computational costs, we truncate article text to the first 2,000 characters³.

2.2 LLM Classification: Definition, Prompt, Implementation

We use [Caldara and Iacoviello \(2022\)](#)’s definition of geopolitical risk, which encompasses wars, escalation of existing events, terrorist attacks, and diplomatic tensions. It covers current risks and events, not historical retrospectives, fictional depictions, or metaphorical uses of geopolitical terminology.

We use the following system prompt to instruct the LLM to evaluate articles:

CLASSIFICATION PROMPT:

You will be given a news article. Classify the article’s assessment of geopolitical risk based only on what the article states or strongly implies.

Geopolitical risk is defined as the threat, realization, and escalation of adverse events associated with Wars (initiation of new conflicts); Escalation of existing wars; Major Terrorist attacks; Tensions between nations or involving cross-border political actors that affect international relations.

Assign a geopolitical risk score from 0.0 to 1.0 based on the following scale:

0.0--0.2: No mention of geopolitical risks.

0.2--0.4: Mentions of minor tensions, diplomatic disputes, or isolated incidents with limited escalation risk.

0.4--0.6: Discussion of significant tensions, ongoing regional conflicts, or moderate escalation risk; some concern about war or terrorism.

0.6--0.8: Substantial discussion of major war initiation/escalation risks, active

² We designed our query around words that align with the definition of geopolitical risk, even if the query does not strictly maximize recall per article retrieved. It is possible to design a query of the same length that captures a larger share of the articles the LLM scores as geopolitical. Such a query selects words such as *insistence*, *devaluation*, and *petroleum*, which are chosen to fit past coverage and need not perform as well on future events.

³ Due to the inverted pyramid structure of journalism where the most important information is presented first, the truncation should capture the essential geopolitical content for the vast majority of articles. As we show in Section 4.3, our baseline scores based on truncated text are nearly identical to those obtained using the full article text.

terrorism threats, or high likelihood of significant escalation.
 0.8--1.0: Extensive coverage of imminent or new war, major war escalation, severe terrorism threats, or critical threat to international stability.
 Movies or books about geopolitical risks, stories remembering old events, obituaries, anniversaries of old events should be given 0 unless they explicitly mention current risks.
 Note: Purely domestic political events (elections, protests, internal policy debates) should score 0.0 unless they have implications for international tensions or cross-border conflicts.

As the prompt illustrates, we use zero-shot prompting where the model receives explicit criteria and a scoring scale but no labeled examples⁴. The scale from 0.0 to 1.0 allows the model to rate articles with varying intensities of geopolitical risk rather than binary classification.

We set the LLM’s temperature parameter to zero, which minimizes stochastic variation in responses across runs. We use structured output to return model output in JSON format, making extraction and processing of scores more efficient.⁵

For each article, we construct a request consisting of the system prompt and the article’s textual content (the headline and first 2,000 characters of the article). All scores fall within the specified 0.0–1.0 range.

2.3 Index Construction

We construct the AI-GPR index by aggregating individual article scores at daily frequency. For each date t , we compute:

$$\text{AI-GPR}_t = \frac{1}{\bar{S}} \times \frac{1}{A_t} \sum_{i=1}^{N_t} S_{it} \quad (1)$$

where S_{it} represents the geopolitical risk score for article i at time t , N_t is the total number of scored articles published on date t , A_t is the total number of articles published on date t across all three newspapers (not just those matching geopolitical keywords), and $\bar{S}$ is a normalization constant chosen such that the index has a mean of 100 over the 1985–2019 period. Dividing by A_t accounts for variation in newspaper output over time and is adopted from [Caldara and Iacoviello](#)

⁴ One-shot or few-shot prompting would require embedding one or more example articles in every request; at the scale of approximately 4.6 million articles, even a single example would roughly double input token costs. The zero-shot approach keeps prompt length short and cost proportional to the number of articles, while the detailed instruction set substitutes for examples by giving the model precise, auditable criteria.

⁵ Structured output is particularly valuable for classification tasks with multiple output dimensions, such as the second-stage classifiers introduced in Section 5.

(2022) for consistency.⁶

The index thus measures the intensity-weighted share of newspaper coverage devoted to geopolitical risk. The numerator sums the risk intensity of each article while the denominator scales by total newspaper output. In Appendix A.2 we show that the index is robust to the main construction choices, particularly whether article scores enter on the intensive or the extensive margin. We verify that the index is not mechanically driven by long-run drift in the scored share N_t/A_t .

Because the index is daily, it records short-lived spikes and dates events precisely, which matters for event studies and for measuring the immediate market reaction to geopolitical developments.

2.4 Computational Considerations

A practical concern for LLM-based approaches is computational cost. At current pricing, processing one article through GPT-4o mini costs approximately \$0.0001 for our truncated text length. For our sample of approximately 4.6 million articles from 1960–2025, the total computational cost is approximately \$450.

Processing time is also manageable. Scoring one article takes roughly one second. With parallelization across five API calls, we can process the full corpus of 4.6 million articles in approximately 250 hours.⁷

3 The AI-GPR Index

Figure 1 displays the AI-GPR index over the full sample, with labels marking the major geopolitical episodes of the past six decades.

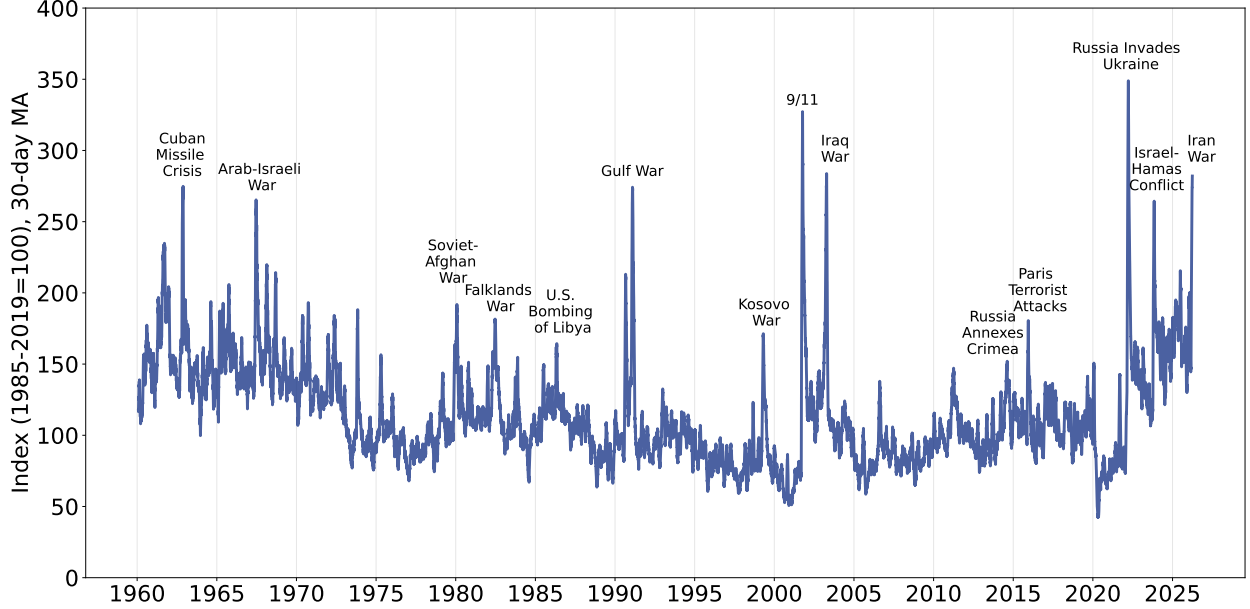
3.1 Summary Statistics

Table 1 presents summary statistics for the AI-GPR index alongside the keyword-based GPR index over the 1960–2025 period as a benchmark. The keyword-based index applies the Caldara and Iacoviello (2022) methodology—which classifies articles using proximity searches of geopolitical and risk-related terms—to the same three newspapers used for the AI-GPR (New York Times, Washington Post, and Chicago Tribune). Caldara and Iacoviello (2022) publish two versions of the index: a historical index based on three newspapers that extends back to 1900, and a recent index based on ten newspapers that begins in 1985. Our version is more comparable with the

⁶ The denominator A_t counts articles containing at least seven common words (“the,” “be,” “to,” “of,” “and,” “at,” and “in”). This choice filters out non-editorial content such as photos with a brief caption, classified advertisements, table-of-contents entries, stock quotes, sports results, and brief notices, and serves as a normalization for trends in newspaper article volume over the years. The total number of articles containing all these words is about 8.5 million. Were we to count every object tagged as an article, the total would be almost twice as large.

⁷ Larger models, such as GPT-5-mini, cost more and take longer to run, but may become feasible as models grow cheaper.

Figure 1: THE AI-GPR INDEX



Notes: The AI-GPR index plotted as a 30-day moving average, from January 1960 through March 31, 2026. The index sums LLM-assigned geopolitical risk scores (0–1) across all articles each day, normalized by total newspaper article count, and is scaled to a mean of 100 over 1985–2019. Regularly updated data are available on the companion webpage. Source: authors’ calculations using ProQuest TDM Studio data.

three-newspaper version, where we sample from the same three newspapers as the AI-GPR back to 1960. The full search string is reported in Appendix A.5. Both indices are normalized so that the mean equals 100 over 1985–2019.

As Table 1 shows, the AI-GPR index is markedly smoother than the original GPR, with a lower standard deviation and a thinner right tail. Its first percentile sits well above zero, and no day records zero geopolitical content, because even quiet days carry some low-level geopolitical discussion. The AI-GPR is also more persistent, with a higher 90-day autocorrelation, because geopolitical conditions evolve smoothly rather than spiking and reverting.⁸

3.2 Comparison with the Original GPR Index

Figure 2 plots the AI-GPR alongside the original GPR index at monthly frequency. The two series co-move closely but diverge in several periods.

In the 1960s–1970s, the AI-GPR index is elevated relative to the original GPR, perhaps because the LLM reads descriptions of geopolitical tension that use words outside the keyword dictionary.

⁸ We measure persistence as the autocorrelation of each index’s 90-day moving average: specifically, the correlation between the smoothed series on day t and on day $t - 90$. This statistic captures slow-moving variation in geopolitical conditions rather than day-to-day volatility, and is consistent with the definition reported in Table 1.

Table 1: AI-GPR Index: Summary Statistics

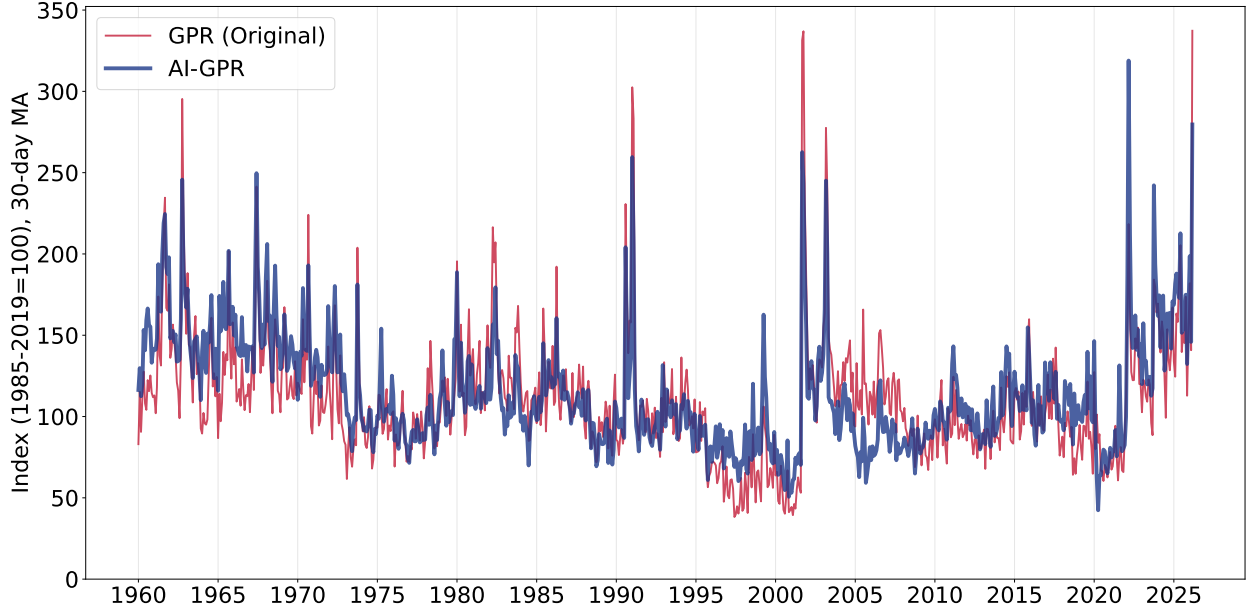
Statistic	AI-GPR	GPR (Original)
Total newspaper articles	8,514,822	8,514,822
Articles sampled	4,571,496	—
Articles scored positive (> 0)	1,166,083	262,096
Standard Deviation	48.52	60.55
Skewness	1.64	1.97
1st Percentile	41.97	18.14
Median	105.46	99.15
99th Percentile	266.95	300.16
90-day Autocorrelation	0.73	0.62
Correlation with Original GPR	0.69	1.00
Positive Articles Share (%)	15.00	3.28
Positive Among Sampled (%)	26.16	3.28

Notes: Summary statistics computed over the 1960–2025 sample. The “GPR (Original)” column is the historical GPR index of [Caldara and Iacoviello \(2022\)](#), computed on the same three newspapers as the AI-GPR over 1960–2025. Both indices are normalized to a mean of 100 over 1985–2019. The 90-day autocorrelation is the correlation between the 90-day moving average on date t and on date $t - 90$. Positive Articles Share (%) is the share of all newspaper articles classified as containing geopolitical risk: for the AI-GPR, articles receiving a positive score; for the original GPR, articles matching the keyword proximity search. Positive Among Sampled (%) is the average daily share of positive articles among those in the queried sample: for the AI-GPR, the share of articles sent to the LLM that receive a score > 0 ; for the original GPR, this coincides with Positive Articles Share (%) since there is no separate sampling stage. The first three rows report totals over the full sample: total newspaper articles (the normalization denominator), articles sampled (sent to the LLM for scoring; the original GPR has no separate sampling stage), and articles scored positive (AI-GPR score > 0 , or a keyword match for the original GPR).

The Cold War-era tensions, the Vietnam War, and the Yom Kippur War all register prominently. In the 1990s, the AI-GPR index is higher during the Bosnian War (1992–1995), suggesting that the LLM captures the sustained geopolitical significance of this conflict more than keyword counting. In the 2000s, both indices spike sharply after the September 11 attacks and during the Iraq War. The AI-GPR declines somewhat faster after the initial Iraq War spike, consistent with the LLM distinguishing articles that cover active conflict from articles that mention the Iraq war in other contexts. From 2010 onward, the indices track each other closely, though the AI-GPR is somewhat more elevated in recent years, perhaps because it captures emerging risks (e.g., cyberwarfare, economic coercion) that do not always trigger keyword matches.

Consistent with [Carlson and Dell \(2025\)](#), the share of articles discussing geopolitical risks is higher under the AI-GPR than in the original GPR tabulations. As shown in Table 1, the AI-GPR classifies 15 percent of all articles as mentioning geopolitical risks, compared to just 3.3 percent

Figure 2: AI-GPR AND ORIGINAL GPR INDEX



Notes: The AI-GPR index (blue, thick) and the original GPR index (red, thin), both plotted at monthly frequency. The original GPR here is the three-newspaper version of the [Caldara and Iacoviello \(2022\)](#) index—their historical index, which extends back to 1900—computed on the same three newspapers as the AI-GPR; the comparison is with this version rather than their ten-newspaper index, which begins in 1985. Both indices are normalized to a mean of 100 over 1985–2019. The AI-GPR index sums LLM-assigned geopolitical risk scores across all articles each day, normalized by total newspaper article count, then averaged to monthly frequency. The original GPR counts articles matching specific keyword proximity searches. Source: authors’ calculations using ProQuest TDM Studio data. Last observation: March 2026.

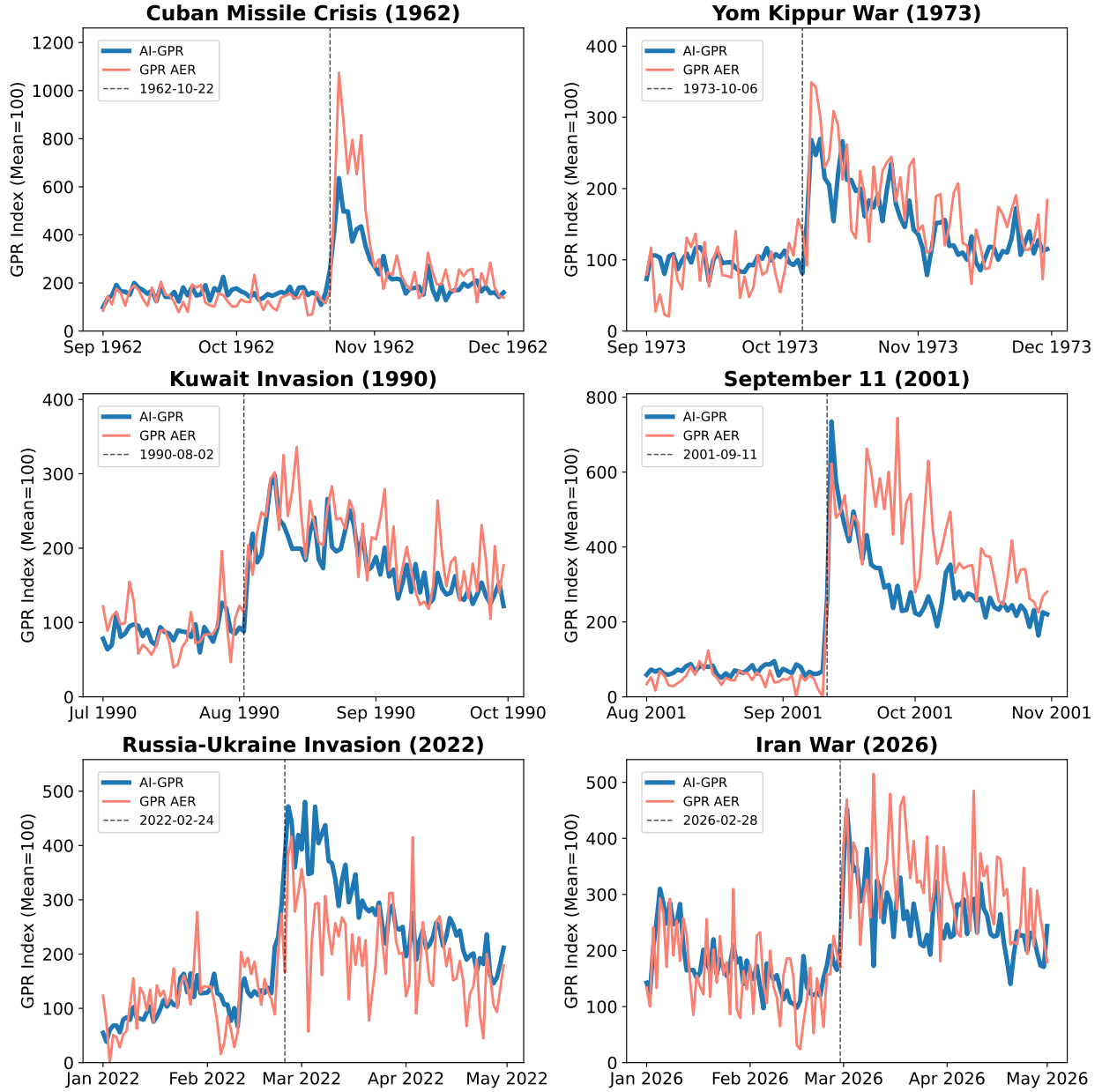
under the original, keyword-based approach.⁹ The two indices nevertheless remain substantially correlated and track each other closely around major events, suggesting that while the keyword approach may understate the share of geopolitically relevant newspaper articles, it still captures the same peaks and troughs.

3.3 Performance on Historical Events

Figure 3 compares the daily AI-GPR and original GPR around six major geopolitical events: the Cuban Missile Crisis, the Yom Kippur War, Iraq’s invasion of Kuwait, the September 11 attacks, Russia’s invasion of Ukraine, and the 2026 U.S. military intervention in Iran. Both series are adjusted for day-of-the-week effects. Both indices spike at the onset of each event, but in every episode the AI-GPR is less volatile from one day to the next. A likely reason is that the LLM recognizes geopolitical risk however a journalist happens to describe it, whereas the keyword index

⁹ Among articles sent to the LLM, 26.2 percent receive a positive score.

Figure 3: AI-GPR vs. ORIGINAL GPR AROUND KEY GEOPOLITICAL EVENTS



Notes: Each panel shows the daily AI-GPR index (blue, thick) and original GPR index (salmon, thin) around a major geopolitical event. Vertical dashed lines mark the event date. Both indices are at daily frequency (no smoothing) and normalized to a mean of 100 over 1985–2019. Both daily indexes are adjusted for day-of-the-week effects using trailing one-year averages of multiplicative day-of-the-week dummies. Source: authors' calculations.

counts an article only when specific terms appear, so it moves with day-to-day changes in word choice. We view this lower noise as one of the strengths of the AI-GPR, especially for financial applications that rely on high-frequency data.

4 Validation and Robustness

This section contrasts the LLM classifier with keyword-based selection, validates the scores against human coding, quantifies the remaining sources of measurement error, and checks that the index is insensitive to the choice of language model.

4.1 LLM versus Keyword-Based Classification

Our LLM-based approach differs from traditional keyword-based classification methods in several ways. [Caldara and Iacoviello \(2022\)](#) classify articles by searching for specific combinations of words from predefined categories. For example, an article is classified as relevant to “war threats” if it contains a geopolitical term (“war,” “conflict,” “hostilities”) within a specified proximity of a risk term (“threat,” “danger,” “crisis”). To reduce false positives, the original GPR methodology employs exclusion words like “movie,” “anniversary,” and “obituary” that disqualify an article from classification even if it contains the requisite geopolitical keywords.

This exclusion strategy addresses some common sources of false positives but remains imperfect. Articles about war films may not use the word “movie” and still discuss fictional geopolitical content. Historical retrospectives may avoid words like “anniversary” while still focusing on past events rather than current events. Moreover, the binary nature of their exclusion methodology (article is either included or excluded entirely) cannot handle articles that legitimately discuss both current geopolitical risks and historical context.

The LLM handles these cases directly. An article that mentions a war film in passing while analyzing current military tensions receives a positive score, whereas an article reviewing a war movie receives a zero score.

Keyword-based approaches face three further limitations. First, keyword matching is context-insensitive: it cannot distinguish whether geopolitical content is central or peripheral to an article, so an article titled “The Price of War in Ukraine” receives the same binary classification as one mentioning “the war on inflation” in a subordinate clause. Keyword-based matching weighs every occurrence of a geopolitical term equally, so it tends to overvalue the geopolitical content of articles that happen to contain such terms. Second, the approach is terminology-dependent, requiring exhaustive keyword lists; articles discussing “armed confrontation” or “military standoff” may be missed if these exact phrases are not in the dictionary or if the proximity requirements are not satisfied. No query can account for all the geopolitical terms used in newspapers, so keyword-based approaches will always miss some articles. Third, binary classification cannot differentiate the intensity or centrality of geopolitical content. The LLM addresses all three: it evaluates whether geopolitical content is central to the article, uses context to distinguish current risks from historical references, recognizes relevant content regardless of terminology, and provides graduated scores reflecting severity.

The tradeoff is reduced transparency. While keyword lists and exclusion rules are fully auditable, LLM-based classification relies on complex neural networks and is effectively a black box. We limit this in two ways. The prompt provides explicit criteria, and we set the temperature of the model to zero to minimize output randomness. The model’s decisions can then be examined by reviewing its classifications on validation samples, which we do next.

4.2 Validation Against Human Judgment

To assess whether the model systematically over- or under-scores articles relative to a human reader, we hand-scored 2,447 articles drawn from all three newspapers, applying the same 0-to-1 scoring rubric given to the LLM. The sample is stratified: it covers articles the keyword filter never retrieved as well as those it did, so the audit speaks to the whole corpus rather than to the articles the model happened to see, and we report rates weighted by the corpus share of each stratum. Each article is scored by an independent human coder and a large subset by a second, which lets us benchmark the model against a human reader and track how closely the human coders agree with one another.¹⁰ Taking the consensus of the human coders as the reference, the machine scores track it closely: their correlation is 0.85, and the two agree on whether an article carries any geopolitical risk in over 90 percent of cases. This exceeds the agreement the human coders reach with one another, who correlate at 0.78 on the double-coded articles, so the model reads geopolitical risk at least as consistently as a second human would. The AI-GPR is a continuous score, so classification-error rates summarize this agreement only coarsely. They do allow a comparison with keyword selection on the same articles. Among the articles the model scores above zero, 13 percent are scored zero by a human coder. The corresponding share for the original [Caldara and Iacoviello](#) keyword query is 16 percent.¹¹ A smaller share of what the AI-GPR flags, then, consists of articles a human reader judges to carry no geopolitical risk. Appendix [A.3](#) reports the full three-way comparison; the model’s and the humans’ average scores are nearly identical, so the classifier introduces no aggregate bias in levels. The pattern we find—a large recall gain over keyword selection at no cost in precision—mirrors what [Hartley \(2025\)](#) documents for the Economic Policy Uncertainty index of [Baker, Bloom, and Davis \(2016\)](#). There, language-model classifiers evaluated on the original human audit raise recall from 45 to 82 percent while precision also improves, which suggests that the advantage of semantic over dictionary classification is a general feature of news-based indexes.

Because the two scores rarely diverge by more than one step on the scale, the disagreements are few enough to read individually, and doing so helps interpret the machine index. Where the

¹⁰ Full details of the audit, such as the inter-rater agreement, the cross-tabulation of human and machine scores, a taxonomy of the disagreements, and illustrative cases, are reported in Appendix [A.3](#).

¹¹ This rate is the share of flagged articles (those receiving a positive score) that the human scores as zero, the quantity comparable to the figure [Caldara and Iacoviello](#) report, which is likewise defined over the articles their keyword search selects. We compute it over individual article-coder ratings rather than the coders’ consensus, so that an article one coder scores as zero and another as positive is not averaged into a positive benchmark.

model scores above the human, two patterns recur: it sometimes reads cross-border economic and financial disputes—tariffs, currency crises, frayed trade ties—and episodes of purely domestic unrest as geopolitical risk, even though the rubric reserves the concept for war, terrorism, and military tension. Its errors of timing are roughly offsetting: it sometimes scores retrospective coverage as though the events were current, and sometimes misses the fresh aftermath of an attack. At the top of the scale the model is conservative, rarely using the highest gradations, so it modestly understates the intensity of the most extreme episodes. A further set of disagreements arises where the human credited context that an article only implies, and so reflects ambiguity in the coding rubric as much as model error. On balance, the audit supports treating the AI-GPR scores as a reliable proxy for human coding.

[Carlson and Dell \(2025\)](#) propose a statistical correction for indices built from text classifiers. They show that keyword- or model-based classifiers produce biased proxies for the true concept of interest, and that this bias can be removed by combining classifier predictions with a human-labeled validation sample that covers the full span of the data. Applied to the GPR index of [Caldara and Iacoviello](#), they find that the keyword-based series underestimates geopolitical risk once the correction is applied. A practical requirement of this approach is a validation sample large and representative enough to pin down the bias across the span of the data. Our LLM approach instead reduces this bias at the classification stage, by replacing keyword matching with a model that reads the article, and produces a similarly elevated series directly. The two approaches are complementary: the LLM lowers classifier error, so the residual bias their framework targets is smaller, and their correction could still be applied to the AI-GPR as a further refinement. Two features of our audit would matter for such an application. Their framework requires a validation sample that is representative of the corpus (missing at random), whereas our audit panel deliberately oversamples articles the model scores as geopolitical, so only its random component, or a suitably reweighted version, would qualify. The correction also inherits the noise in the human labels: our double-coding shows that human coders disagree with one another somewhat more than the model disagrees with their consensus, a bound on annotation quality that their framework takes as given.

4.3 Measurement Error

Four potential sources of measurement error affect the AI-GPR index: model stochasticity, text truncation, attenuation from the two-stage filtering process, and hindsight knowledge embedded in the language model.

Model stochasticity. Even at temperature zero, LLMs are not deterministic and may produce slightly different outputs across calls. To quantify this, we run GPT-4o mini three times on the same subset of 1,050 articles. Scores are nearly identical across the three runs—the vast majority

of articles receive exactly the same score—so classification noise from model stochasticity is negligible. Raising the temperature to 0.5 and 1.0 barely changes this: within- and cross-temperature agreement remains high and mean scores are nearly the same.¹² The multi-model comparison in Section 4.4 gives a further check: scores are highly correlated across four OpenAI models and an open-weight Llama model, implying that classification is driven by article content rather than idiosyncratic model behavior.

Text length. Our baseline classifier truncates article text at 2,000 characters to reduce processing time and cost. To verify that this truncation does not materially affect classification, we re-score the same articles using their complete text. Truncated and full-text scores are highly correlated and identical for most articles. Full-text scores are slightly higher on average, as longer articles carry additional geopolitical content beyond the opening paragraphs. The ranking of articles is nearly unchanged.¹³

First-stage attenuation. The keyword pre-filter could introduce measurement error by missing geopolitically relevant articles or by including irrelevant ones. We assess the false-omission rate—the share of the articles the filter discards that are in fact geopolitically relevant—by applying the LLM classifier to a random sample of ten thousand articles, drawn from all three newspapers, that did not match any of our search keywords. Only 0.6 percent received a positive geopolitical risk score. The few that did scored low: none reached the midpoint of the scale, and the discarded articles taken together would add almost nothing to the index. This check relies on the model’s own judgment of what counts as relevant, so two of us also read 400 of the discarded articles, each scoring them independently on the rubric of Section 2.2 without seeing the model’s score. One of us found geopolitical content in two of them, the other in twelve, and neither scored any article above 0.4. Pooling the two readings, fewer than two percent of the discarded articles carry any geopolitical content in a human’s judgment, and none carries much. The model, which scored these same articles in the false-omission check, flags one of the 400: the article both of us flagged, and at the same 0.2. On the articles the filter discards, then, the model sides with the stricter of the two human readings. These 400 articles form one of the three strata of the human audit discussed in Appendix A.3. False positives from our keyword filter are handled by the LLM scoring step: articles that match keywords but do not discuss geopolitical risk receive a score of zero and do not contribute to the index.

¹² Across the three temperature-zero runs the pairwise correlation is 0.99, with at least 98 percent of articles scored identically and a mean absolute difference below 0.004. Cross-temperature agreement between temperature 0.0 and 1.0 is 94.1 percent, with a mean absolute difference of 0.012. Correlation and agreement against the human-annotated sample remain very high and similar across the temperature 0.5 and 1.0 runs.

¹³ The correlation between truncated and full-text scores is 0.93, with 86.7 percent of articles scored identically; the mean score rises from 0.129 to 0.153 with full text.

Hindsight knowledge. Because GPT-4o mini is trained on text through a recent cutoff, it scores historical articles already knowing how the events they describe unfolded—that the Cuban Missile Crisis was resolved without war, or how the response to September 11 played out. Were the model to lean on this knowledge, it could systematically misstate the risk perceived by a contemporaneous reader. To probe this, we re-score the articles surrounding four major events—the Cuban Missile Crisis, the 1990 invasion of Kuwait, the September 11 attacks, and Russia’s 2022 invasion of Ukraine—with a modified prompt that fixes each article’s own publication date as the present (“Today is {date}”), supplies the date and headline, and explicitly instructs the model that it has no knowledge of events occurring after that date. The resulting series is nearly indistinguishable from the baseline. The date-anchored scores run marginally higher for the Cuban Missile Crisis and September 11, the two crises that resolved without the feared catastrophe, and marginally lower for Kuwait and Russia-Ukraine, so the small differences do not point in one direction. Appendix A.4 plots the comparison and reproduces the full prompt. Re-scoring the randomly audited articles with the same prompt gives the same answer: the scores barely change, and the small average shift is again slightly upward. The caveat is that telling a model it lacks knowledge of the future does not erase that knowledge from its parameters, so the test shows that invoking or denying such knowledge barely moves the index, not that the knowledge is absent.

4.4 Robustness to AI Model Choice

To assess whether our choice of LLM affects the resulting index, we classify a sample of 2,050 articles using four OpenAI models—GPT-4o mini (our baseline), GPT-4o, GPT-5-mini, and GPT-5—and an open-weight model, Llama 3.1 8B Instruct. All models receive the same prompt. We set temperature to zero for the models that support it; GPT-5-mini and GPT-5 do not expose an adjustable temperature, so they run at their default setting with reasoning effort set to high.¹⁴ GPT-4o mini’s scores are highly correlated with those of every alternative, and the share of articles classified as containing geopolitical risk is similar across all of them.¹⁵ This agreement holds across models spanning two generations, different model sizes, and both proprietary and open-weight model families, suggesting that the AI-GPR index is not sensitive to the specific model used. Given that GPT-4o mini is substantially cheaper and produces highly correlated scores, we adopt it as our baseline for the full corpus. The agreement with an open-weight model also speaks to reproducibility: proprietary models are periodically deprecated, a concern Carlson and Dell (2025) raise for AI-derived measures generally, and the Llama results indicate that the index can be reconstructed with a model whose weights are permanently and publicly available.

¹⁴ For Llama, we run meta-llama/Llama-3.1-8B-Instruct locally with 8-bit quantization, applying the model’s standard instruct chat template, and decode greedily for near-deterministic output.

¹⁵ The pairwise correlations between GPT-4o mini and the alternatives are 0.90 (GPT-4o), 0.87 (GPT-5-mini), 0.86 (GPT-5), and 0.91 (Llama); within the GPT-5 family, GPT-5-mini and GPT-5 correlate at 0.94. The positive-classification share ranges from 26 to 33 percent across models.

5 The Anatomy of Geopolitical Risk

We apply further LLM classifiers to measure three additional dimensions of geopolitical risk: threats and acts, oil, and country-level and bilateral risk. The threats-and-acts measure identifies whether geopolitical risk is anticipated or realized. The oil measure captures the intensity of oil-supply disruptions driven by geopolitical events. The country and bilateral measures track which countries are involved in geopolitical events and the directed relationships between them. Each is produced by applying an additional classification layer to the underlying articles.¹⁶

5.1 Threats and Acts

A key innovation of [Caldara and Iacoviello \(2022\)](#) was to separate geopolitical risk into *threats*—adverse events that are anticipated, feared, or warned of but have not yet materialized—and *acts*—events that have actually occurred or are underway. The distinction is economically meaningful, because anticipated and realized risk need not move the economy in the same way, as our analysis of stock returns in [Section 6.1](#) makes clear. We build an analogous decomposition of the AI-GPR index with a second LLM classification layer that reads each article and judges the temporal nature of the risk it describes.

We apply the classifier to every article that receives a positive AI-GPR score, passing the same text supplied to the baseline classifier to GPT-4o mini at temperature zero.¹⁷ The model places each article in one of three categories. An article is a *threat* if it centers on risks, warnings, or tensions that have not yet come to pass. It is an *act* if it reports an adverse event that has occurred, such as an attack, invasion, or coup. It is *both* if it discusses a realized event together with the risk of further escalation. Because articles in the *both* category speak to each dimension, we count them toward both subindexes, so that the threat series aggregates the articles labeled *threat* or *both* and the act series aggregates those labeled *act* or *both*.¹⁸ Each subindex is then built exactly like the headline AI-GPR index: we sum the geopolitical risk scores of its articles, divide by total newspaper volume, and rescale the series to average 100 over 1985–2019.

[Figure 4](#) plots the two subindexes at monthly frequency, and they move differently. Acts spike sharply around realized conflicts—the 1991 Gulf War, the September 11 attacks, the 2003 Iraq War, and Russia’s 2022 invasion of Ukraine—while threats run higher during the early Cold War and other spells of elevated but unrealized tension, when coverage is dominated by warnings and

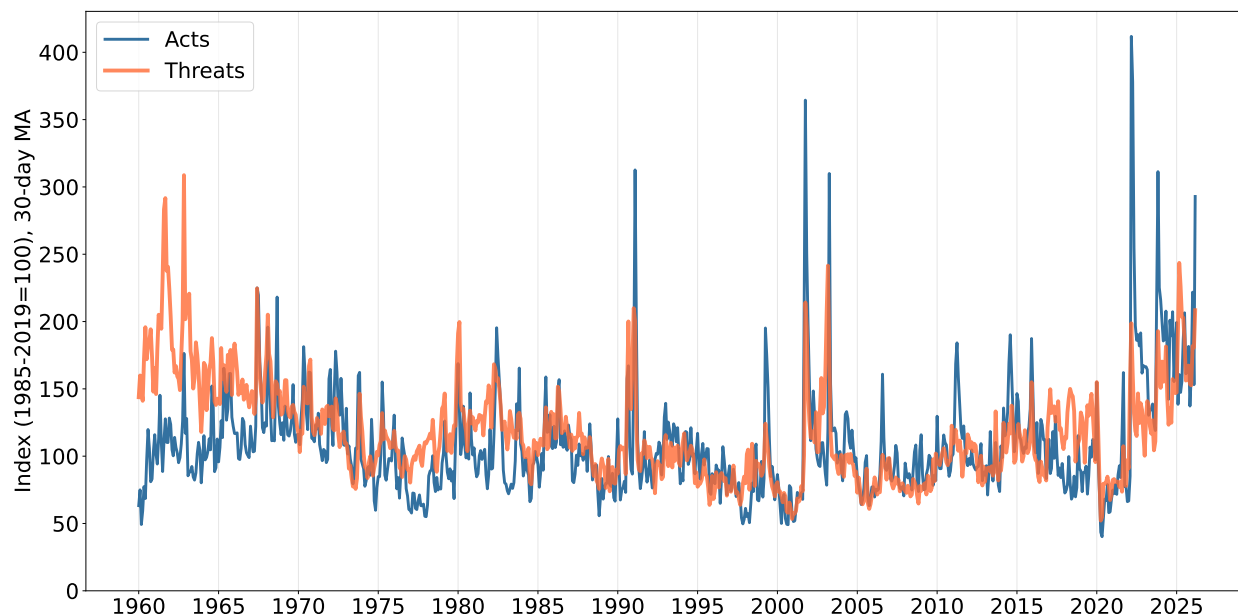
¹⁶ Every subindex in this section is normalized by the same denominator as the headline index in [equation \(1\)](#): the relevant geopolitical risk scores are summed and divided by A_t , the total count of newspaper articles published on date t across all three papers (not just those in the relevant category). Using this common denominator throughout—rather than, for instance, the number of articles within each category—keeps all of the indexes on the same scale as the aggregate AI-GPR index and ensures their levels remain directly comparable.

¹⁷ As in the baseline scoring step, the model reads the first 2,000 characters of each article. The full prompt is reproduced in [Appendix A.7](#).

¹⁸ About 15 percent of classified articles fall in the *both* category, with roughly 48 percent labeled threats and 37 percent acts.

brinkmanship rather than open conflict.

Figure 4: GEOPOLITICAL THREATS AND ACTS



Notes: The threats subindex (orange, thick) and acts subindex (blue, thin) of the AI-GPR index, plotted at monthly frequency. A second LLM classification layer assigns each geopolitically relevant article (AI-GPR score > 0) to “threat,” “act,” or “both”; articles labeled “both” contribute to each series. Series are indexed so that each series averages 100 over 1985–2019. Source: authors’ calculations using ProQuest TDM Studio data. Last observation: March 2026.

5.2 Oil Markets

We construct a second subindex about oil supply disruptions driven by geopolitical events using a two-step classification process similar to Section 5.1. In the first step, we restrict attention to articles receiving a GPR score above 0.5 and, to economize on classification costs, retain those mentioning at least one oil- or energy-related keyword; articles without such keywords are recorded as carrying no oil-supply risk. In the second step, we apply a second LLM prompt to the retained articles.¹⁹ This prompt asks GPT-4o mini to determine whether the article discusses an oil or energy supply disruption caused by a geopolitical event and, if so, which geographic region(s) are affected from a predefined list: Middle East, Russia, USA, Venezuela, North Africa, West Africa, Central Asia, North Sea, Canada, Mexico, Latin America, Southeast Asia, and China.²⁰

¹⁹ To improve classification accuracy, the second-stage classifiers pass the first 3,000 characters of each article to GPT-4o mini, compared to 2,000 characters for the base GPR scoring step.

²⁰ The full classification prompt is located in Appendix A.7.

For each date t , we construct the oil disruption ratio:

$$\text{Oil GPR}_t = \frac{\sum_{i \in D_t} S_{it}}{A_t} \quad (2)$$

where D_t is the set of articles on date t classified as discussing a geopolitical oil disruption, S_{it} is the article’s GPR score (from the first classification layer), and A_t is the total number of newspaper articles published on date t . The numerator weights each disruption article by its GPR intensity, so that an article extensively covering an oil crisis (score near 1.0) contributes more than one with a passing mention (score near 0.5). The denominator normalizes by total publishing volume, analogously to the construction of the AI-GPR index. Regional oil disruption indices are constructed similarly, restricting D_t to articles classified as affecting a particular region. We also split the index along the threat-act dimension of Section 5.1: keeping only the disruption articles that the classifier labels as threats, or as both threats and acts, gives an Oil GPR threats index, and the complementary selection gives an acts index. The threats index isolates the risk of a future disruption from the disruption itself, which is what makes it useful as an instrument for oil price movements in Section 6.2.

Figure 5 plots the Oil GPR index alongside oil prices. The index closely tracks the major geopolitical oil supply events: the 1973 Arab oil embargo, the 1979 Iranian Revolution, the 1990–91 Gulf War, and the Russia-Ukraine conflict and associated sanctions from 2022 onward. The co-movement between the Oil GPR index and oil prices confirms that the LLM-based classification is capturing the geopolitical episodes most associated with supply disruptions. A regional decomposition of the Oil GPR index by geography—showing the shifting dominance of the Middle East through the 1990s, Russia after 2014, and Venezuela during its political crises—is provided in Figure A.3 of the appendix.

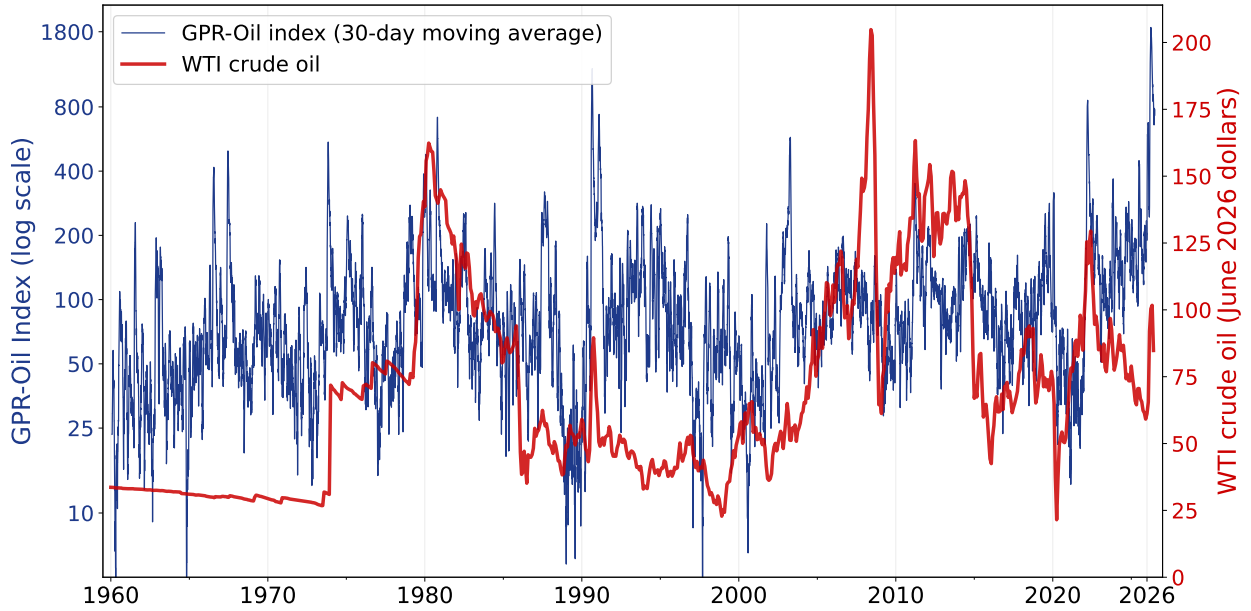
Of all articles with a GPR score above 0.5, about 13 percent contain oil-related keywords and are sent to the second classification layer. Of these, roughly three-quarters are classified as discussing a geopolitical oil supply disruption. The Middle East accounts for nearly two-thirds of all disruption mentions, consistent with its dominant role in geopolitical oil risk throughout the sample period. Russia is the second most frequently mentioned region at 14 percent, with its share rising sharply after 2014.

5.3 Country Risk and Bilateral Risk

Our final set of measures focuses on country-level and bilateral geopolitical risk.

The country subindex measures the intensity of geopolitical events associated with a particular country. To construct the subindex, we parse each article through a classification layer to identify which countries are involved in the geopolitical event and their corresponding actor roles. We

Figure 5: GPR OIL INDEX AND REAL OIL PRICES



Notes: The dark blue line (left axis, logarithmic) is the Oil GPR index (30-day moving average of the daily series). The red line (right axis) is the monthly real WTI crude oil price, in June 2026 dollars. Last observation is June 30, 2026.

aggregate the AI-GPR scores for each country across all roles and scale to the overall index.²¹ Figure 6 presents monthly series for the United States, Russia, China, and the United Kingdom—the four countries most often featured in newspaper coverage of geopolitical risks. The companion webpage includes time series of geopolitical risk for 200 countries.

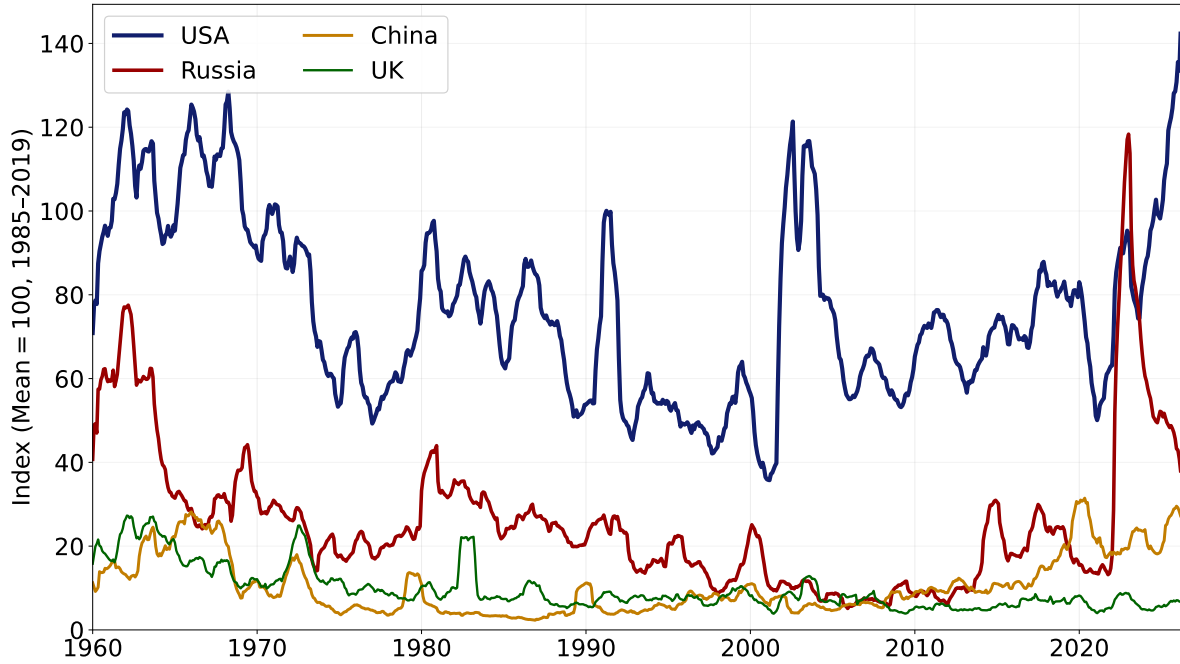
The United States is present throughout the sample, as a party to or a target of conflicts worldwide.²² Russia registers two distinct elevated periods: the Cold War era through the late 1980s, and the post-2014 period following the annexation of Crimea and the full-scale invasion of Ukraine in 2022. China’s GPR contribution rises steadily from the late 1990s onward, consistent with its growing global presence and the intensification of geopolitical competition in the Indo-Pacific. Country-level decompositions of this kind extend the country sub-indices of [Caldara and Iacoviello \(2022\)](#), who constructed keyword-based series for selected nations. The LLM approach enables a more precise construction of sub-indices across all countries mentioned in the news without requiring pre-specified keyword lists.

We build on the country subindex to create the bilateral index. This index tracks the level of geopolitical risk between initiator-respondent country pairs. We construct bilateral indices for the top 1,200 country pairs as the sum of AI-GPR sentiment scores in articles where the first country

²¹ The full country classifier prompt is in Appendix A.7.

²² The large coverage of U.S.-related events also reflects our choice of U.S. newspapers. For measures built from non-U.S. news sources, see [Ambrocio et al. \(2025\)](#), [Andres-Escayola et al. \(2022\)](#), and [Bondarenko et al. \(2024\)](#).

Figure 6: AI-GPR BY COUNTRY



Notes: Monthly AI-GPR contribution for the four most prominent countries across all actor roles (initiator, respondent, spillover), smoothed with a 12-month moving average. Series are scaled to the overall AI-GPR index (mean = 100, 1985–2019). Source: authors’ calculations using the second-stage LLM country classifier applied to articles with positive GPR score. Last observation: March 2026.

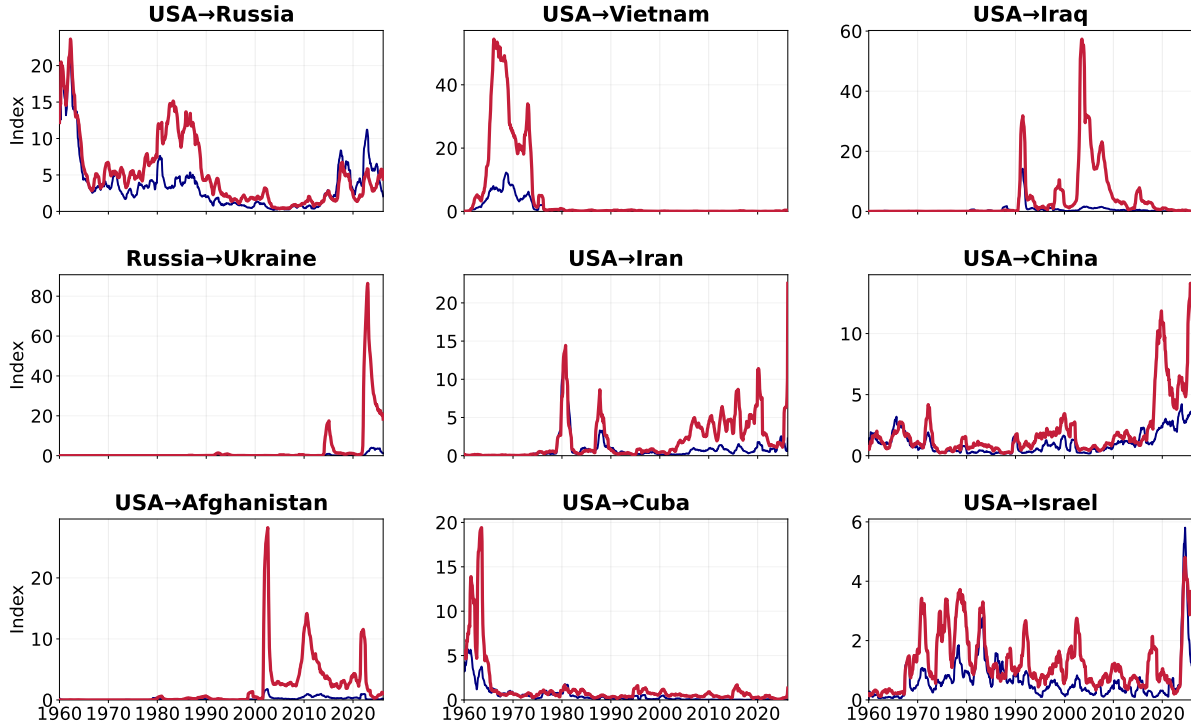
is classified as initiator and the second as respondent, so that the USA→Russia and Russia→USA series are distinct.²³

Keyword methods cannot be used to build this index. They can tell that Russia and the United States both appear in an article, but not whether Russia is acting on the United States, whether the United States is responding to Russian pressure, or whether the two countries merely appear in adjacent paragraphs about unrelated topics. The LLM classifier records which country initiates the geopolitical action and which is the target.

Figure 7 presents the nine country pairs with the highest average bilateral intensity, each shown in both directions: the thick red line is the more intense direction (named in the panel title) and the thin blue line is its reverse. The United States is the dominant initiator in eight of

²³ Each directed-pair index is scaled analogously to the country-level indices: the raw sentiment sum for the pair is multiplied by the ratio of the overall AI-GPR index to the total coded sentiment in that month. This scaling is algebraically equivalent to dividing the pair’s sentiment sum by the same total article count A_t used for the benchmark index (and applying the benchmark’s rescaling constant), so the bilateral series share the aggregate index’s denominator and scale rather than being renormalized by the smaller set of articles involving each pair. Because an article in which one country acts against multiple respondents contributes its full sentiment score to each initiator–respondent pair (and vice versa), the bilateral indices do not in general sum to the aggregate index.

Figure 7: AI-GPR BILATERAL INDEXES: TOP COUNTRY PAIRS, BOTH DIRECTIONS



Notes: Each panel shows both directed bilateral AI-GPR sub-indices for a country pair, smoothed with a 12-month moving average. The thick red line is the more intense direction—initiator → respondent, as named in the panel title—and the thin blue line is the reverse direction. Each directed sub-index aggregates AI-GPR sentiment scores from articles in which the first country is classified as initiator and the second as respondent, scaled to the overall AI-GPR index. The nine pairs have the highest average intensity in their dominant direction over the sample. Last observation: March 2026. Source: authors’ calculations using the second-stage LLM country classifier applied to articles with positive GPR score.

the nine pairs, reflecting both its central role in postwar geopolitics and the US-centric newspaper sources underlying the index. The pairs span the Cold War confrontation with the Soviet Union (USA→Russia, USA→Cuba), the major US military interventions of the postwar era (USA→Vietnam, USA→Iraq, USA→Afghanistan), ongoing great-power and regional tensions (USA→China, USA→Iran, USA→Israel), and Russia’s 2022 invasion of Ukraine (Russia→Ukraine). The directional split is itself informative: when a large power confronts a smaller one—USA→Iraq, Russia→Ukraine, USA→Afghanistan—coverage casts one country overwhelmingly as the initiator and the reverse series barely registers; between comparably sized powers, such as the United States and Russia or the United States and China, initiation is attributed far more evenly across the two directions.

The same classification layer also labels each article by the type of geopolitical event it describes—among them military conflict, diplomatic tension, terrorism, civil war, nuclear threat, coup or

regime change, and sanctions.²⁴ These categories track familiar historical patterns. Military conflict peaks around the Korean and Vietnam Wars, the Gulf War, and Russia’s 2022 invasion of Ukraine. Terrorism rises sharply after the September 11 attacks. Sanctions become more prominent after 2014, as economic coercion becomes a favored geopolitical instrument.

6 Applications: Geopolitical Risk in Financial Markets, Oil Markets, and Trade

We present a series of applications relating the three dimensions of geopolitical risk to financial markets, oil, and trade. First, we examine how geopolitical threats and acts are priced in the stock market using the threats and acts subindexes. We then study the macroeconomic response to oil shocks identified with an instrument constructed from the Oil GPR index. Lastly, we examine how bilateral geopolitical risk relates to international trade using our bilateral GPR index.

6.1 Financial Markets and Geopolitical Threats and Acts

A large literature studies whether geopolitical events depress stock markets. We revisit this question using the AI-GPR index and daily data on U.S. stock returns from 1960 to March 2026.

We merge the AI-GPR index with daily excess market returns.²⁵ Since stock markets are closed on weekends and holidays while the AI-GPR is computed for every calendar day, we construct a trading-day GPR measure as follows. If the market is closed for k consecutive calendar days before trading day t , we assign to day t the average of the AI-GPR over those $k + 1$ days (including day t itself), so that geopolitical information accumulating over non-trading days is correctly attributed to the next trading day. We then work with the first difference of this series, standardized by its full-sample standard deviation. All GPR variables in the regressions below are standardized in this way. We estimate the regression:

$$r_t^e = \alpha + \beta \Delta \text{GPR}_t + \varepsilon_t \quad (3)$$

where r_t^e is the excess market return (in percent). Table 2 reports the results estimated on data collapsed to weekly frequency, where stock market returns are cumulated weekly and the GPR change is computed week over week. The relationship is negative and economically meaningful: a one-standard-deviation increase in the AI-GPR is associated with a decline of about fourteen basis

²⁴ The event-type classification prompt is listed in Appendix A.7.

²⁵ Through November 28, 2025, we use data from the Fama-French daily factors dataset. See https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. Beginning December 1, 2025, when Fama-French daily factors are not yet available, we extend the return series using the daily excess return on the S&P 500 index from Haver Analytics, computed as the daily simple return on the index minus the daily-compounding equivalent of the three-month Treasury bill rate.

Table 2: Weekly Stock Returns and Geopolitical Risk

	Aggregate GPR		Threats and Acts	
	OLS (1)	Decomposition (2)	OLS (3)	Decomposition (4)
ΔGPR_t	−0.136** (0.057)			
$\Delta\text{GPR}_{t, \text{threats}}$			−0.081** (0.037)	
$\Delta\text{GPR}_{t, \text{acts}}$			−0.084 (0.061)	
Persistent $\widehat{\Delta\text{GPR}}_t$		−0.271** (0.134)		
Persistent $\widehat{\Delta\text{GPR}}_{t, \text{threats}}$				−0.220** (0.103)
Persistent $\widehat{\Delta\text{GPR}}_{t, \text{acts}}$				−0.063 (0.131)
Shock $\hat{u}_t$		−0.119* (0.062)		
Shock $\hat{u}_{t, \text{threats}}$				−0.052 (0.044)
Shock $\hat{u}_{t, \text{acts}}$				−0.090** (0.042)
Constant	0.137*** (0.037)	0.138*** (0.037)	0.136*** (0.037)	0.138*** (0.037)
Observations	3,452	3,448	3,452	3,448
R^2	0.004	0.004	0.004	0.004
Sample	1960–2026	1960–2026	1960–2026	1960–2026

Notes: Weekly regressions of excess market returns (r_t^e , in percent) on standardized changes in the AI-GPR index. Columns (1)–(2) use the AI-GPR index; columns (3)–(4) use its threats and acts subindexes. Weekly stock returns are the sum of daily excess returns within the week, and the weekly GPR variable is the change in the within-week average of the trading-day GPR measure, standardized over the full sample. Columns (1) and (3) report OLS regressions of r_t^e on the GPR change(s). Columns (2) and (4) decompose each GPR change into a persistent component (fitted values from an AR(4) model) and a shock component (residuals). Excess market returns are from the Fama-French daily factors dataset through November 28, 2025, extended with Haver SP500 excess returns (SP500 simple return minus the daily-compounded 3-month T-bill rate) from December 1, 2025 onward. OLS columns report heteroskedasticity-robust standard errors in parentheses; decomposition columns report pairs-bootstrap standard errors (1,000 replications). The estimation sample runs through March 2026. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

points in the weekly market return.

The OLS estimate in equation (3) conflates predictable and unpredictable movements in GPR. To disentangle these, we decompose geopolitical risk into anticipated and unanticipated components. We estimate an AR(4) model for the standardized GPR change:

$$\Delta\text{GPR}_t = \phi_0 + \sum_{j=1}^4 \phi_j \Delta\text{GPR}_{t-j} + u_t \quad (4)$$

and decompose ΔGPR_t into the persistent (predictable) component $\widehat{\Delta\text{GPR}}_t$ and the shock (unpre-

dictable) component $\hat{u}_t$. We then estimate:²⁶

$$r_t^e = \alpha + \beta_1 \widehat{\Delta\text{GPR}}_t + \beta_2 \hat{u}_t + \varepsilon_t \quad (5)$$

We estimate both regressions—the level specification in equation (3) and the anticipated vs. unanticipated decomposition in equation (5)—twice: first using the aggregate AI-GPR index, and then replacing it with the threat and act subindexes entered jointly. The two pairs of regressions map onto the four columns of Table 2—the aggregate index in columns (1) and (2), and its threat-and-act refinement in columns (3) and (4).

For the aggregate index, the predictable, persistent component of geopolitical risk depresses returns far more than the unexpected component—its effect is more than twice as large—and both are statistically significant. A sudden geopolitical event like the September 11 attacks generates an immediate negative return, but the persistence of elevated risk in the weeks that follow is what depresses returns most. Markets re-price as risk persists and enters the outlook, not only on the day of the event.

The gap between the muted aggregate OLS estimate and the much larger effect of the persistent component suggests that a simple linear regression understates the role of geopolitical risk, attenuated by the high-frequency noise of weekly shocks, which are large in magnitude but transient in their effect on asset prices.²⁷

Separating the index into its threat and act components adds detail. In levels, the effect of *threats* on returns is precisely estimated, while realized *acts*—though comparable in magnitude—are too noisy to register as statistically significant. The decomposition into anticipated and unanticipated movements shows why. Threats affect returns through their predictable part: as a threat persists and enters the outlook, markets re-price gradually, so it is the slow-moving, anticipated component of threat risk that weighs on returns. Acts work the other way around. Foreseeable events are already in prices, so only the *unanticipated* occurrence of an act moves markets. This asymmetry also explains why acts look insignificant in the simple regression: pooling a surprise effect with a negligible anticipated one pushes the average toward zero.

The improved day-to-day precision of the AI-GPR index is central to these results. When we repeat the same analysis using the keyword-based GPR index²⁸, the results are qualitatively similar

²⁶ This approach follows the empirical tradition of Barro (1979), who decomposes macroeconomic variables into anticipated and unanticipated components to test whether agents respond differently to predictable trends versus stochastic innovations.

²⁷ This result echoes the findings in Gonçalves et al. (2025), who examine the pricing of geopolitical risk using stock market returns. They use Caldara and Iacoviello’s decomposition of geopolitical risk into forward-looking threats (GPT) and realized acts (GPA), and find that only the threat component is consistently priced across assets and forecasts equity premia, whereas realized acts have a much weaker and less stable link to risk premia.

²⁸ To keep this comparison on the same newspapers and time period as the AI-GPR, we reconstruct Caldara and Iacoviello’s (2022) keyword methodology ourselves on the same three newspapers, extending it back to 1960; the publicly available daily GPR series covers ten newspapers and begins only in 1985. See Section A.5.

but economically and statistically weaker: the contemporaneous relationship with returns is roughly a third smaller and only marginally significant, and in the decomposition the persistent component barely clears conventional significance while the shock component loses it altogether. Table A.6 in Appendix A.6 reports these regressions in full. These differences illustrate how measurement imprecision in the keyword-based index may attenuate the estimated effects. Keyword-based methods misclassify articles at higher rates, so the resulting index contains more variation that is not geopolitically relevant, biasing coefficients toward zero in a standard errors-in-variables fashion.

6.2 Oil Market Shocks and their Macroeconomic Effects

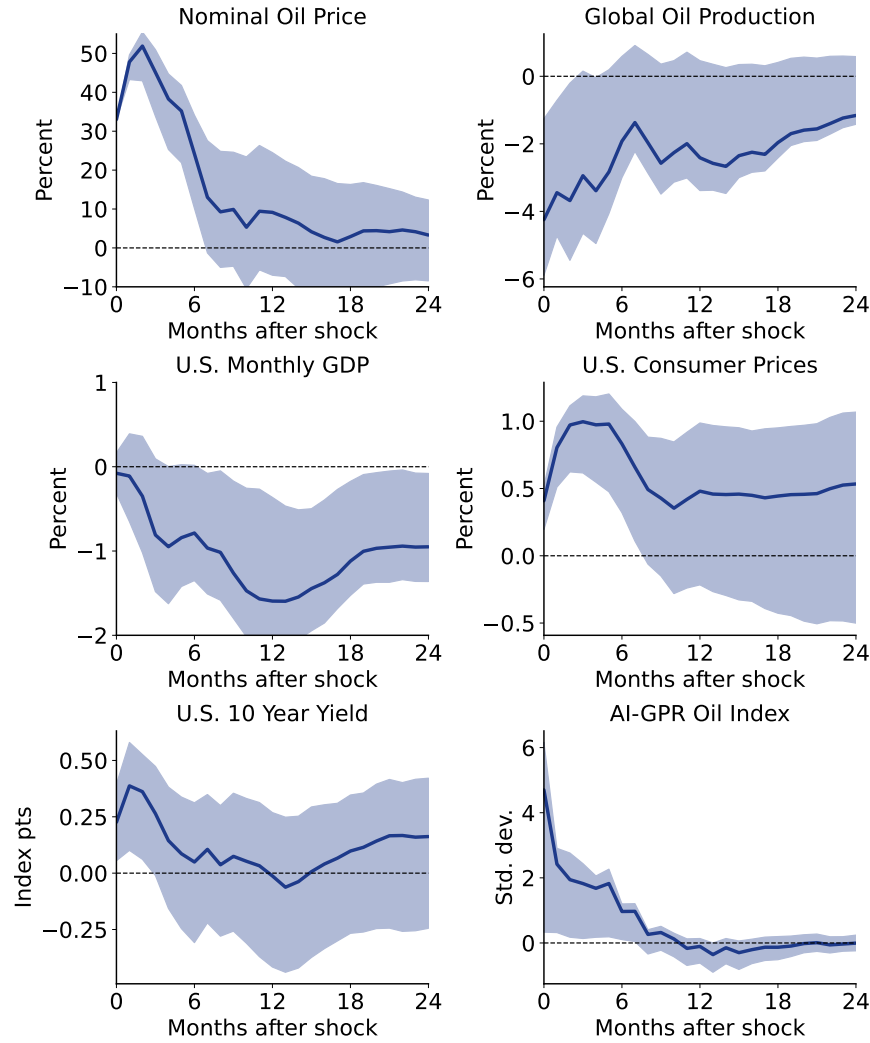
The Oil GPR index can help isolate oil price movements driven by geopolitical supply disruptions, as distinct from demand-driven fluctuations or weather-related supply shocks. We borrow this idea from Verduzco-Bustos and Zanetti (2026), who identify geopolitically driven oil shocks from oil price changes on days with large jumps in the keyword-based GPR threats index; we replace their news series with the Oil GPR index, which isolates geopolitical news related to energy supply. We estimate a six-variable proxy structural VAR (Mertens and Ravn, 2013; Stock and Watson, 2012) at monthly frequency with 12 lags, from January 1985 to May 2026.²⁹ The system includes the nominal U.S. crude oil price, global oil production, U.S. monthly GDP (the Brave-Butters-Kelley index, Brave et al. 2019), the U.S. consumer price index, the 10-year Treasury yield, and the Oil GPR threats index described below.

We build the instrument by constructing the threats component of the Oil GPR index: we keep only the oil-disruption articles that the classifier of Section 5.1 labels as threats or as both threats and acts, and aggregate them as in equation (2). Next, we flag days on which the index is more than one standard deviation above its mean. Verduzco-Bustos and Zanetti (2026) flag only extreme daily jumps in the keyword-based index; a lower bar is appropriate here because the LLM-based index measures geopolitical news with less noise than keyword counts (Section 4.3). Finally, we move each flagged day back by one day and measure the log change in the daily WTI spot price from the last trading day before that date to the first trading day after it; for events around weekends, the window spans the weekend. The monthly instrument is the sum of these price changes. It is zero in months without flagged days. The instrument is a strong one: it enters the first stage with an F -statistic of 45, well above the conventional threshold, so the responses below are not subject to the weak-instrument problems that often affect proxy VARs. We compute confidence bands with the moving block bootstrap of Jentsch and Lunsford (2022). Appendix A.9 reports the data sources, the variable transformations, the block length and the number of draws.

Figure 8 presents the impulse responses to a geopolitical oil supply shock, normalized to raise

²⁹ Daily oil price quotes start in 1986, so the instrument is zero for every month of 1985. A proxy VAR does not require the instrument to be observed in every period, and with 12 lags the first year of data serves as initial conditions.

Figure 8: IMPULSE RESPONSES TO A GEOPOLITICAL OIL PRICE SHOCK



Notes: Impulse responses to a geopolitical oil supply shock identified with a proxy structural VAR. The shock is normalized to raise oil prices by 33 log points on impact. Shaded areas are 80 percent confidence intervals from a moving block bootstrap.

oil prices by about one third on impact. Oil prices keep rising for two months and then fall back gradually. The Oil GPR threats index jumps on impact and stays elevated for about six months. Oil production falls on impact and recovers gradually. Consumer prices rise, peaking at about 1 percent above baseline, and 10-year yields rise. U.S. GDP declines gradually and is about one and a half percent lower a year after the shock, reflecting the contractionary effect of higher energy costs as well as the broader disruptions that accompany geopolitical events affecting the oil market.

6.3 Bilateral Geopolitical Shocks and International Trade

Is higher geopolitical risk between two countries accompanied by visible economic effects? We merge our bilateral AI-GPR index with the CEPII Gravity database (Conte et al., 2023), which assembles UN Comtrade flows into a harmonized dataset of annual bilateral trade. Our variable of interest is total two-way real merchandise trade for each country pair, obtained by summing the two directed flows (in US dollars) and deflating using the US GDP deflator. The trade data cover 1962 to 2020, the last year in the current release of the database.³⁰

Trade flows are likely to respond to geopolitical news with some delay, as trading relationships take time to build and redirect. Fan et al. (2026), who construct a bilateral geopolitical alignment score from a large-scale classification of political events, find that trade adjusts to changes in alignment gradually over several years. A static regression of trade on contemporaneous or once-lagged risk may thus miss these effects. To trace the response of trade to bilateral geopolitical risk at each horizon, we estimate:

$$y_{od,t+h} - y_{od,t-1} = \alpha_{od,h} + \alpha_{o,t,h} + \alpha_{d,t,h} + \gamma_h Z_{od,t} + \mathbf{d}_h \mathbf{X}_{od,t} + \varepsilon_{od,t+h}, \quad (6)$$

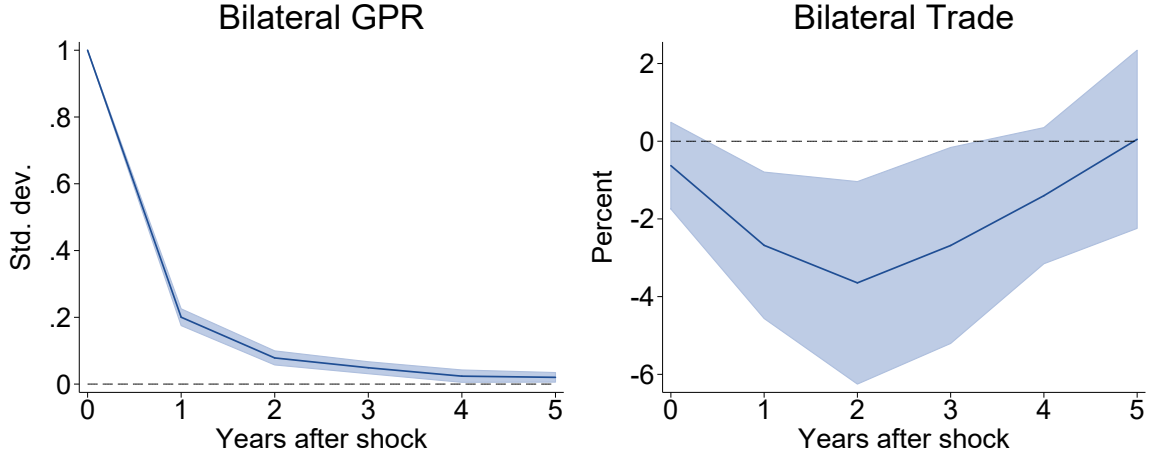
where $y_{od,t} = 100 \times \log X_{od,t}$, $X_{od,t}$ is total two-way trade between the pair, and $Z_{od,t}$ is the bilateral AI-GPR index, standardized within country pair so that it measures how far a relationship has moved relative to its own norm rather than its level in index points. The goal is to estimate, for each horizon h , the coefficient γ_h , which measures the dynamic effect on bilateral trade of a one-standard-deviation increase in bilateral geopolitical risk. Since y is 100 times log trade, each γ_h is the percent response of trade at horizon $t+h$ relative to year $t-1$. The regression includes country-pair fixed effects (α_{od}) and separate origin-by-year and destination-by-year dummies ($\alpha_{o,t}$ and $\alpha_{d,t}$). Controls $\mathbf{X}_{od,t}$ include three lags of bilateral GPR, three lags of $y_{od,t}$, and a dummy for whether the pair is party to a regional trade agreement. Conditioning on past GPR is what makes $Z_{od,t}$ a shock rather than a level. The projection runs from $h=0$ onward: with three lags of both variables on the right-hand side, the pre-shock path is already absorbed by the controls.³¹ Standard errors are clustered three ways, by origin, destination and year, since geopolitical events move many country pairs in the same year. We estimate the same projection, with the same right-hand side, replacing trade with bilateral GPR on the left.

Figure 9 plots the two responses. The left panel shows that a typical bilateral shock is short-lived: the increase in risk has unwound within a year. The right panel shows that trade responds more slowly. It barely moves in the year of the shock, then declines over the following two years, reaching a trough of roughly 3.5 percent below baseline before gradually recovering. Trade is

³⁰ Comtrade records each flow twice, as reported by the exporter and by the importer; we use the importer's report and fall back on the exporter's where it is missing.

³¹ The variable is taken from the WTO's Regional Trade Agreements database, supplemented by Thierry Mayer, as distributed in Conte et al. (2023).

Figure 9: RESPONSE OF BILATERAL GPR AND TRADE TO A BILATERAL GPR SHOCK



Notes: Responses to a one-standard-deviation increase in bilateral GPR, where the standard deviation is computed within country pair. Both panels plot the value at $t + h$ minus the value at $t - 1$. The left panel is bilateral GPR, in standard deviations. The right panel plots the coefficients γ_h from equation (6), the percent response of total two-way bilateral trade. Shaded areas are 90 percent confidence intervals, with standard errors clustered by origin, destination and year.

still falling at horizons where the risk itself has largely passed, which is what one would expect if contracts and supply relationships take time to redirect. This transitory decline in trade is consistent with the evidence in [Fan et al. \(2026\)](#) for a transitory shock to geopolitical alignment.

7 Conclusion

We have introduced the AI-GPR Index, an improved measure of geopolitical risk that uses large language models to evaluate newspaper content. Our approach maintains the conceptual framework established by [Caldara and Iacoviello \(2022\)](#) while replacing keyword matching with a model that reads each article. This reduces false positives from irrelevant keyword matches and false negatives from relevant articles that use other terminology, and it yields graduated scores that reflect the intensity of geopolitical content. The method could be extended to non-English publications and other text sources.

We illustrated several applications of the index. First, the AI-GPR produces a more precisely estimated negative effect of geopolitical risk on stock returns and shows that markets are especially sensitive to the persistent component of GPR innovations; decomposing the index into threats and acts shows that this persistent effect runs through anticipated threats, while the surprise component of realized acts also weighs on prices. Second, by combining the AI-GPR with a second classification layer, we constructed a historical time series of geopolitical oil supply disruptions by region, showing

the shifting geography of energy-related geopolitical risk over six decades and the macroeconomic effects of geopolitically driven oil shocks. Third, we extract the geopolitical actors in each event and their roles. This yields country indices, which extend the sub-indices of [Caldara and Iacoviello \(2022\)](#), and indices for directed country pairs, which identify who is acting on whom rather than which countries merely appear together. A rise in bilateral GPR is followed by a decline in trade between the two countries.

Our main contribution is demonstrating that LLM-based approaches can be applied at scale to construct consistent historical time series spanning multiple decades. As computational costs decline, this becomes practical for a wide range of text-based measurement in economics, and the approach complements rather than replaces keyword-based methods. Future research could extend it to firm-level or sector-level measures of geopolitical risk exposure, or use the bilateral and country-level indices to study how geopolitical risks transmit across countries and markets.

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A Appendix

A.1 Alternative Versions of AI Model to Classify Articles

We classify a sample of 2,050 articles using four OpenAI models—GPT-4o mini, GPT-4o, GPT-5-mini, and GPT-5—together with the open-weight Llama 3.1 8B Instruct, all with the same prompt. We set temperature to zero for the models that support it; the GPT-5 models do not expose an adjustable temperature and run at their default settings. Table A.1 reports the full pairwise correlation matrix.

Table A.1: Cross-Model Score Correlations

	GPT-4o mini	GPT-4o	GPT-5-mini	GPT-5	Llama
GPT-4o mini	1.00				
GPT-4o	0.87	1.00			
GPT-5-mini	0.87	0.86	1.00		
GPT-5	0.87	0.88	0.94	1.00	
Llama 3.1 8B	0.91	0.85	0.84	0.84	1.00

Notes: Pairwise Pearson correlations of geopolitical risk scores assigned by each model to a sample of 2,050 newspaper articles drawn from all three publications. All models use the same classification prompt, at temperature zero where supported; the GPT-5 models run at their default settings. “Llama” denotes the open-weight Llama 3.1 8B Instruct model.

All pairwise correlations are high, with the closest agreement between GPT-5-mini and GPT-5. The mean score varies modestly across models, reflecting differences in calibration that wash out after normalization.

A.2 Alternative Ways of Constructing the Index

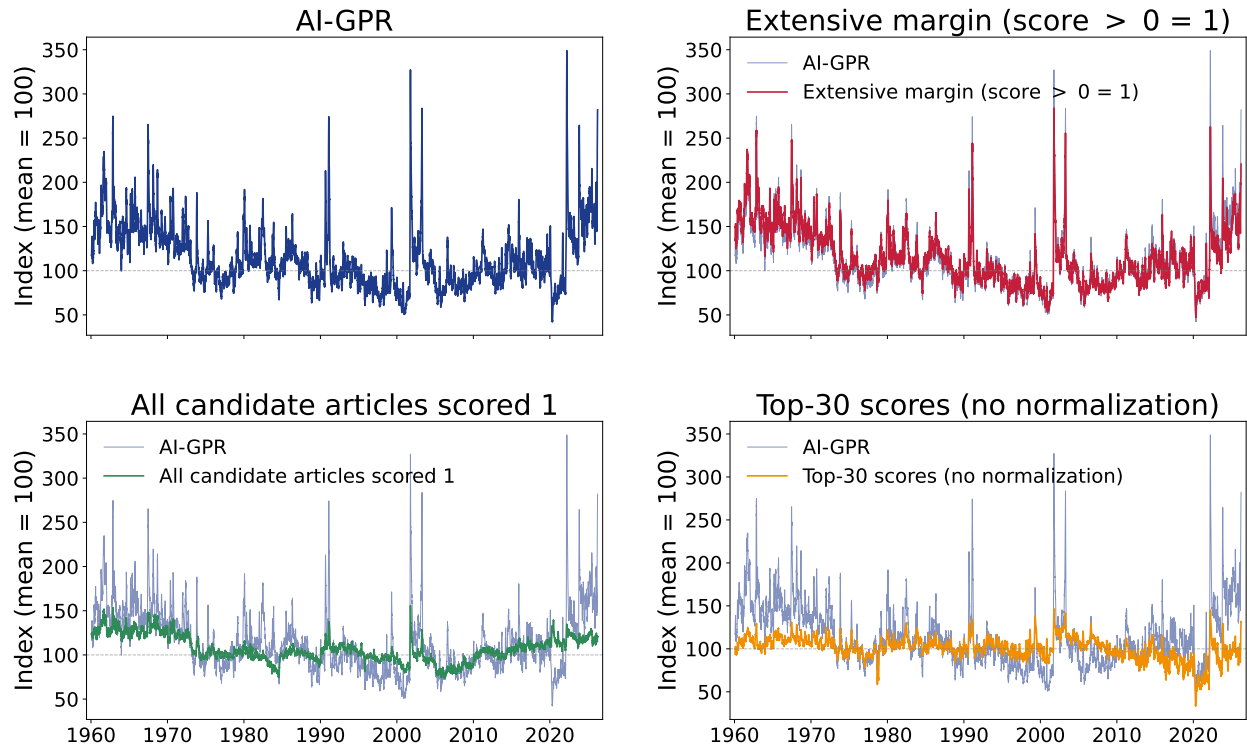
The baseline index in equation (1) makes two choices a reader might dispute: it weights each article by its graded risk score S_{it} (the intensive margin), and it normalizes the score sum by total newspaper output A_t rather than by the number of scored articles N_t . To show the role of these choices, we recompute the index under four alternative constructions, each reindexed to a mean of 100 over 1985–2019: (i) the baseline AI-GPR, $\sum_i S_{it}/A_t$; (ii) an *extensive-margin* version in which every article with $S_{it} > 0$ counts as one, $\#\{i : S_{it} > 0\}/A_t$, discarding intensity; (iii) an *all-candidate* version in which every scored article counts as one regardless of its score, equal to the scored share N_t/A_t ; and (iv) an *unnormalized* version equal to the sum of the thirty highest scores each day, $\sum_{\text{top } 30} S_{it}$, with no denominator.

Figure A.1 plots the four series and Table A.2 reports their pairwise correlations. The baseline and the extensive-margin version are very close (correlation 0.97): the intensity weighting is not what generates the index’s movements, although it emphasizes some of the spikes. Replacing the graded score sum in the numerator with a simple count of scored articles—the all-candidate share N_t/A_t —lowers the correlation to 0.70, and removing the denominator A_t entirely (the unnormalized top-30 sum) gives 0.48 in levels; both nonetheless track the baseline closely in year-on-year changes (0.66 and 0.85, respectively). These construction choices therefore change the index’s high-frequency movements little.

The concern that the index might merely reflect long-run drift in the scored share N_t/A_t —as newspaper composition and the share of topics shift over six decades—is addressed by the third

construction. As Figure A.1 shows, the share of candidate articles is a smooth, slowly drifting series with none of the event spikes that define the geopolitical risk index, and among the four it least resembles the baseline’s sharp, event-driven movements. In addition, the series shows a U-shape that mostly reflects the inverted U-shape in the pattern of total newspaper articles over the sample. The index’s economically relevant variation—the sharp rises around the Gulf War, September 11, the Iraq War, and Russia’s invasion of Ukraine—comes from the geopolitical content of the articles, not from how many articles clear the generous keyword filter.

Figure A.1: ALTERNATIVE VERSIONS OF THE INDEX



Notes: Four alternative constructions of the AI-GPR index, each shown as a 30-day moving average and reindexed to a mean of 100 over 1985–2019. *AI-GPR* is the baseline $\sum_i S_{it}/A_t$; *Extensive margin* counts each article with $S_{it} > 0$ as one; *All candidate articles scored 1* counts every scored article as one, equal to N_t/A_t ; *Top-30 scores* sums the thirty highest scores each day with no normalization. Source: authors’ calculations. Last observation: March 2026.

A.3 Additional Details on the Human Audit

This appendix gives the full results of the human validation summarized in Section 4.2. We hand-score 2,447 articles drawn from the *New York Times*, the *Washington Post*, and the *Chicago Tribune* between January 1960 and December 2025. We use the same prompt and guidelines that the model uses to score the articles. 2,101 articles were scored by two annotators, and we take the consensus of the available coders as the human score and compare it with the LLM’s score.

The audit is stratified rather than a simple random draw, because geopolitical articles are a small minority of the corpus and a proportional sample of the corpus would contain unrelated articles not flagged by our filters. Table A.3 gives the design. Stratum C, articles that are flagged

Table A.2: Correlations Across Index Constructions

	(1)	(2)	(3)	(4)
(1) AI-GPR (baseline)	1.00			
(2) Extensive margin	0.97	1.00		
(3) All candidate (N_t/A_t)	0.70	0.75	1.00	
(4) Top-30 (unnormalized)	0.48	0.41	0.08	1.00

Notes: Pairwise Pearson correlations of the four index constructions defined in the text, computed on the daily series (reindexed to a mean of 100 over 1985–2019) over 1960–2026. A further variant not shown in the figure, normalizing by scored articles N_t instead of total articles A_t —the average score among scored articles, $\sum_i S_{it}/N_t$ —yields a series correlated 0.89 with the baseline.

by keyword filter and which the model scored above zero, is where humans can disagree on an article the LLM scored positively; stratum B, flagged but scored zero, is where the model’s misses would show up; and stratum A, never retrieved at all, is where the keyword filter’s misses would live. The three strata make up the corpus, so weighting each by its population share recovers rates for the full sample of 8.5 million articles. The weights differ substantially—an audited article in stratum A stands for five times as many corpus articles as one in stratum C—so the pooled, unweighted rates describe the audited panel and not the corpus, and we report both.

Table A.3: Design of the Human Audit

Stratum	Share of corpus (%)	Articles audited	Corpus articles each represents
A. No keyword match	46.3	400	9,893
B. Keyword match, AI score 0	40.0	1,451	2,357
C. Keyword match, AI score > 0	13.7	596	1,970
Total	100.0	2,447	3,495

Notes: The three strata partition the 8.5 million articles published in the three newspapers over 1960–2025. Stratum A is articles the keyword pre-filter never sent to the LLM, drawn from the same pool as the false-omission check of Section 4.3; strata B and C are articles it did send, split by whether the model scored them above zero. The last column is the stratum population divided by the number of articles audited in it, that is, the weight each audited article carries when rates are computed for the corpus.

The human coders agree with one another imperfectly: on the double-coded articles in strata B and C their scores correlate at 0.78 and concur on whether an article carries any geopolitical risk in about 92 percent of cases. Even careful readers applying the same rubric make different judgment calls, which sets a benchmark for any classifier. Against the human consensus, the machine scores correlate at 0.85—above the coders’ agreement with each other—agree on the presence of any geopolitical risk in about 90 percent of cases, and have a mean absolute difference of about 0.05, roughly the spacing of a single step on the scale. Table A.4 reports the full cross-tabulation; the off-diagonal mass sits almost entirely on the first off-diagonal, so disagreements are overwhelmingly one-step judgment calls at a rubric boundary, such as whether an ongoing regional conflict constitutes “significant tensions” (0.4) or “major war escalation risks” (0.6). The model’s and the humans’ average scores are essentially identical (both 0.122), so the article-level disagreements are offsetting

and introduce no aggregate bias in levels.

The model’s disagreements with the humans are, moreover, concentrated in the articles the humans themselves find ambiguous. On the double-coded articles where the two coders assign identical scores, the model matches their common score exactly in about 90 percent of cases, with a mean absolute deviation of 0.02; where the coders disagree with each other, the model’s mean absolute deviation from the consensus rises to 0.14. Model error thus tracks genuine ambiguity in the articles rather than occurring at random, the same pattern [Hartley \(2025\)](#) documents for policy-uncertainty classification, where model misclassification falls by half as human-coder certainty rises.

Table A.4: Human vs. AI Score Cross-Tabulation

Human score	AI score				
	0.0	0.2	0.4	0.6	0.8
0.0	1366	60	23	3	0
0.2	43	34	47	6	0
0.4	39	60	140	101	0
0.6	3	0	21	73	5
0.8	0	1	3	15	4

Notes: Cross-tabulation of the human consensus score (rounded to the nearest 0.2) and the GPT-4o mini score, for the 2,047 audited articles in strata B and C of [Table A.3](#). The 400 stratum-A articles are left out because the pre-filter never sent them to the model, so they carry no machine score in the index. Both human coders scored 388 of the 400 zero, one coder found geopolitical content in two and the other in twelve, and no article was scored above 0.4. Scored for the false-omission check of [Section 4.3](#), the model flags one of them, the article both coders flag, at the same 0.2. These articles enter the corpus-weighted rates of [Table A.5](#). Rows are human scores, columns are model scores. The model never assigned a score above 0.8. Off-diagonal mass is concentrated on the first off-diagonal, that is, one-step disagreements.

Scoring the same audited articles with both classifiers also allows a like-for-like comparison against the human labels, free of the differences in sample, rubric, and ground truth that complicate cross-study figures. [Table A.5](#) reports this three-way comparison, pooling the individual article–coder ratings in the two keyword-retrieved strata (3,748 pairs, with each double-coded article entering once per coder) and applying the original [Caldara and Iacoviello \(2022\)](#) keyword query—exclusion words included—to the same texts. Benchmarking each coder’s rating separately, rather than the consensus mean, ensures that an article one coder scores as zero and another as positive does not count as positive by construction. Within each method’s own set of flagged articles, the share a human coder scores as zero is broadly similar (a false-discovery rate of about 13 percent for the AI-GPR and 16 percent for the keyword query, the latter below the 21 percent [Caldara and Iacoviello](#) report from their own audit), confirming the precision advantage discussed in [Section 4.2](#). Measured against the consensus mean instead, the false-discovery rates are somewhat lower for both methods (9.4 and 15.1 percent), since averaging turns some single-coder zeros into positive benchmarks. The keyword query flags far fewer articles overall. Bringing stratum A back in and reweighting the three strata to their corpus shares, in the right-hand panel of the table, leaves that gap essentially intact: measured against the whole sample of 8.5 million articles rather than the retrieved strata alone, the AI-GPR fails to flag about a quarter of what a human coder scores as geopolitical, the keyword query more than four fifths. Stratum A is the one input the retrieved strata cannot supply, and the two coders who read it differ on how much geopolitical content it holds: one found it in half a percent of the articles, the other in three percent. Taking either coder

alone instead of both moves the AI-GPR’s corpus miss rate by about three percentage points in either direction and the keyword query’s by less than one, so the gap between the two methods does not depend on which reading one takes. The reweighting matters more for the keyword query’s false-discovery rate, which rises by close to two percentage points because its flagged articles are drawn from both of the retrieved strata, while the AI-GPR’s is unchanged because its flagged set is stratum C itself.

Table A.5: Human, AI-GPR, and Keyword GPR Compared

	Retrieved strata (unweighted)			Full corpus (stratum-weighted)		
	Flagged	False-disc.	Miss	Flagged	False-disc.	Miss
Human coders	30.5	—	—	15.6	—	—
AI-GPR	29.2	13.2	16.9	13.7	13.2	23.8
Keyword GPR	6.9	16.3	81.1	3.3	18.1	82.5

Notes: The audit covers all 2,447 articles of Table A.3, stratum A included; the two panels differ in which of them they aggregate and how. The left-hand panel pools the article–coder ratings in the two strata the keyword filter retrieved, each article entering once for every human coder who scored it, 3,748 ratings in total across the *New York Times*, *Washington Post*, and *Chicago Tribune*. It sets stratum A aside because the three strata are sampled at different rates, so an unweighted pooling is interpretable only across strata sampled alike. The right-hand panel brings stratum A back in, where both coders read every article and each rating enters once, and weights the three strata by their corpus shares, so that it describes the full sample of 8.5 million articles. An article is flagged by a method when its score is positive; the keyword row applies the original [Caldara and Iacoviello \(2022\)](#) query, including its exclusion words, to the same texts. The false-discovery rate is the share of a method’s flagged articles that the human scores as zero. The miss rate is the share of articles the human scores as positive that the method fails to flag. The AI-GPR’s false-discovery rate is the same in both panels because its flagged set coincides with stratum C, while the keyword query’s rises under reweighting because its flagged set spans strata B and C.

Reading the disagreeing articles and recording notes on each case, we group the discrepancies into five recurring patterns.

First, the model treats occasional economic and financial tensions between countries as geopolitical risks. Articles on a proposed U.S. tariff on Japanese goods (1971), the financial crisis in Latvia (2009), Iceland’s refusal to repay British and Dutch lenders (2010), and strained Taiwan–China business ties (2000) were scored 0.4 to 0.6 by the model and 0 to 0.2 by us. The rubric defines geopolitical risk around war, terrorism, and military tensions. Trade and financial disputes between friendly nations strain “ties” in a literal sense but rarely carry risk of armed conflict, and a human coder discounts them accordingly. Many of the cases in which the model scores higher than the human are of this type, consistent with the model anchoring on surface vocabulary (“threat,” “retaliation,” “frayed ties”) rather than the military nature of the underlying risk.

Second, the model treats domestic unrest and domestic security as geopolitical risk, even if the original definition emphasizes the interstate nature of such risks. Internal events with security overtones—the 1991 Crown Heights riots, labor strife framed around a general strike, an article on the disciplining of unruly airline passengers before the 2021 inauguration—were scored 0.4 to 0.6 by the model. The prompt explicitly instructs that domestic political events should score zero absent international implications, but the model appears to weight words such as “violence,” “riot,” and “security” heavily even in domestic settings.

Third, the model makes some minor temporal-attribution errors in both directions. The rubric instructs that retrospective coverage of old events should score zero unless current risks are mentioned. In some cases the model fails to recognize aftermath coverage as backward-looking: a 2004 article on legal turmoil in a Detroit terrorism prosecution (events eight months past) received 0.6, and a 2002 piece on routine airport security received 0.4. In other cases it applies the retrospective rule too aggressively: human-interest profiles published days after the 1993 World Trade Center bombing and the 1998 embassy bombings were scored 0 by the model even though the attacks were current events signaling elevated terrorism risk. The model evaluates the 2,000-character snippet in isolation, whereas a human coder knows the article’s date and the surrounding history.

Fourth, the human coder sometimes credits context the snippet only implies. A 1961 wire item on a Navy missile contract was scored 0.2 by us (arms development at the height of the Cold War) and 0 by the model, which is arguably the more faithful reading of the prompt’s instruction to classify based only on what an article states or strongly implies. A brief 2003 item on the capture of Ali Hassan al-Majid scored 0.6 in our coding—it presupposes the ongoing Iraq war—and 0 from the model. Reports on the humanitarian consequences of conflict, such as a 1973 piece on war-induced famine in Cambodia, scored higher in our coding because we treated humanitarian devastation and the implied risk of escalation as risk content while the model adhered to the narrower definition. In these cases the human is more generous than the prompt strictly licenses, so the disagreement reflects the rubric as much as the model.

Fifth, the model occasionally scores the backdrop rather than the subject of an article. When a major conflict serves as background for a story about something else—a 2023 piece on how the war in Ukraine affects livelihoods in a Norwegian border town, a 2017 color piece on the mood in a Beirut neighborhood—the model seems to score the conflict (0.6) while we score the article (0.2). The conflicts are real, but the articles convey little incremental information about war risk. These are occasional instances where a human coder weighs the subject of the article, while the model weighs the most alarming entity mentioned in it.

Finally, the model appears to compress the top of the scale: it rarely uses the highest gradations and almost never scores above 0.8, whereas human coders occasionally do. The compression is mild and affects few observations, but it implies that the machine index understates the relative intensity of the most extreme episodes. Taken together, the audit supports treating the LLM scores as a reliable proxy for careful human coding, with the caveat that the index seems to run slightly hot at the bottom of the scale, and slightly cold at the top.

A.4 Hindsight Robustness Check

This appendix accompanies the hindsight discussion in Section 4.3. To test whether the model’s knowledge of how historical events unfolded contaminates its scores, we re-score the articles around four major events with a date-anchored variant of the baseline prompt. The variant fixes each article’s own publication date as the present, supplies the date, headline, and publisher alongside the article text, and instructs the model that it has no knowledge of anything that happened afterward. The full prompt is reproduced below; its geopolitical risk definition and 0.0–1.0 scoring scale are identical to the baseline prompt of Section 2.2.

```
HINDSIGHT-CHECK PROMPT (system message, prepended with the article’s own date):  
Today is {date}
```

You are a geopolitical risk analyst reading a news article. You possess absolutely zero knowledge of any events that occurred after today. Classify the article’s assessment of geopolitical risk based only on what the article states or strongly implies. [The geopolitical risk definition and the 0.0--1.0 scoring scale follow verbatim from the baseline prompt in Section 2.2.]

User message:
Date: {date}
Title: {title}
Publisher: {publisher}
{article text}

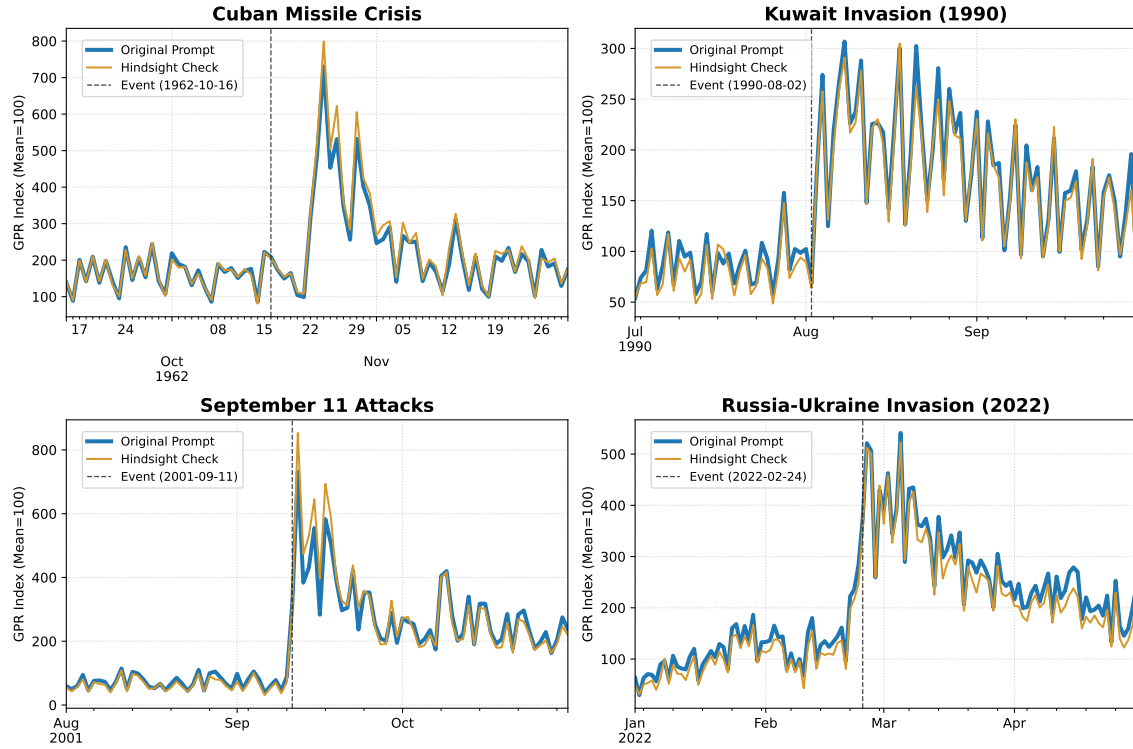
Figure A.2 plots the resulting daily series against the baseline around each event. The two are nearly indistinguishable. On average the hindsight-corrected scores are marginally higher for the Cuban Missile Crisis and September 11—the direction consistent with mild hindsight dampening in the baseline—and marginally lower for Kuwait and Russia-Ukraine, but the differences are very small throughout.

A.5 The Original Caldara and Iacoviello (2022) Keyword-Based GPR Index

As described in Section 3, we compare the AI-GPR index against the original, keyword-based GPR index developed by Caldara and Iacoviello (2022), using the same three newspapers and time period as the AI-GPR. In the original Caldara and Iacoviello (2022) formulation, the search terms are organized into eight categories, each combining a geopolitical term group with a risk/threat term group using either the AND or the N/2 (words must appear within 2 words of each other) proximity operators. The full search string is:

Caldara and Iacoviello (2022) KEYWORD-BASED GPR SEARCH QUERY:
(war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR revolt OR coup OR geopolitical) N/2 (risk* OR warn* OR fear* OR danger* OR threat* OR doubt* OR crisis OR troubl* OR disput* OR concern* OR tension* OR imminen* OR inevitable OR footing OR menace* OR brink OR scare OR peril*)
OR (peace OR truce OR armistice OR treaty OR parley) N/2 (menace* OR reject* OR threat* OR peril* OR boycott* OR disrupt*)
OR (military OR troops OR missile* OR ‘‘arms’’ OR weapon* OR bomb* OR warhead*)
AND (buildup* OR build-up* OR blockad* OR sanction* OR embargo OR quarantine OR ultimatum OR mobiliz*)
OR (‘‘nuclear war’’ OR ‘‘atomic war’’ OR ‘‘nuclear missile*’’ OR ‘‘nuclear bomb*’’ OR ‘‘atomic bomb*’’ OR ‘‘h-bomb*’’ OR ‘‘hydrogen bomb*’’ OR ‘‘nuclear weapon*’’)
AND (risk* OR warn* OR fear* OR danger* OR threat* OR doubt* OR crisis OR troubl* OR disput* OR concern* OR tension* OR imminen* OR inevitable OR footing OR menace* OR brink OR scare OR peril*)
OR (terrorism* OR guerrilla* OR hostage*) N/2 (risk* OR warn* OR fear* OR danger* OR threat* OR doubt* OR crisis OR troubl* OR disput* OR concern* OR tension* OR

Figure A.2: BASELINE AND HINDSIGHT-ANCHORED AI-GPR AROUND KEY EVENTS



Notes: Each panel plots the daily AI-GPR index around a major geopolitical event under the baseline prompt (blue, thick) and under the hindsight-check prompt (orange, thin), which fixes the article’s publication date as the present and instructs the model that it has no knowledge of subsequent events. Both series are scored on the same articles, normalized by total newspaper article count, and scaled to a mean of 100 over 1985–2019. Vertical dashed lines mark the event date. Source: authors’ calculations using ProQuest TDM Studio data.

```
imminen* OR inevitable OR footing OR menace* OR brink OR scare OR peril*)
OR (war OR conflict OR hostilities OR revolution* OR insurrection OR uprising OR
revolt OR coup OR geopolitical) N/2 (begin* OR begun OR began OR outbreak OR
‘broke out’ OR breakout OR start* OR declar* OR proclamation OR launch*)
OR (allie* OR enem* OR foe* OR army OR navy OR aerial OR troops OR rebels OR
insurgen*) N/2 (drive* OR shell* OR advance* OR offensive OR invasion OR invad* OR
clash* OR attack* OR raid* OR launch* OR strike*)
OR (terroris* OR guerrilla* OR hostage*) N/2 (act OR attack OR bomb* OR kill* OR
strike* OR hijack*)
NOT (movie* OR film* OR museum* OR anniversar* OR obituar* OR memorial* OR arts OR
book OR books OR memoir* OR ‘price war’ OR game OR story OR history OR veteran*
OR tribute* OR sport OR music OR racing OR cancer OR ‘real estate’ OR mafia OR
trial OR tax)
```

Table A.6: Weekly Stock Returns: AI-GPR versus the Original Keyword-Based GPR

	AI-GPR		GPR-AER	
	(1) OLS	(2) Decomposition	(3) OLS	(4) Decomposition
ΔGPR_t	-0.136** (0.057)		-0.097* (0.053)	
Persistent $\widehat{\Delta \text{GPR}}_t$		-0.271** (0.134)		-0.239** (0.118)
Shock $\hat{u}_t$		-0.119* (0.062)		-0.072 (0.064)
Constant	0.137*** (0.037)	0.138*** (0.037)	0.137*** (0.037)	0.138*** (0.038)
Observations	3452	3448	3452	3448
R^2	0.004	0.004	0.002	0.003
Sample	1960–2026	1960–2026	1960–2026	1960–2026

Notes: Weekly regressions of excess market returns on standardized changes in GPR, identical specification and sample to Table 2. Columns (1)–(2) use the AI-GPR index; columns (3)–(4) use GPR-AER, the reconstructed three-newspaper Caldara and Iacoviello (2022) keyword index described in Section A.5. OLS columns report heteroskedasticity-robust standard errors in parentheses; decomposition columns report pairs-bootstrap standard errors (1,000 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The resulting index is the ratio of articles matching the proximity search to total articles published on each date, normalized to have a mean of 100 over 1985–2019. This is the “GPR (Original)” series reported throughout the paper.

A.6 Stock Returns Using the Original GPR Index

Section 6.1 reports that repeating the stock-return regressions with the keyword-based GPR index in place of the AI-GPR yields qualitatively similar but weaker results. Table A.6 reports the regressions underlying that comparison in full, alongside the AI-GPR results of Table 2 for reference. Columns (1)–(2) reproduce Table 2’s aggregate-index columns exactly; columns (3)–(4) repeat the same specification with GPR-AER, the reconstructed three-newspaper index described in Section A.5. The aggregate OLS coefficient under GPR-AER is about a third smaller than under AI-GPR and only marginally significant. In the decomposition, the persistent component remains negative and barely clears the 5 percent level, while the shock component loses significance altogether—consistent with the view that the keyword index’s higher misclassification rate attenuates the estimated relationship.

A.7 Prompts for the AI-GPR Subindexes

We use the following prompts to classify articles for the GPR subindexes.

A.7.1 Threats and Acts GPR

The threats and acts classification prompt identifies the nature of geopolitical risk. Relevant articles are labeled based on whether they predominantly discuss anticipated risk (threats), realized risk (acts), or both.

THREATS AND ACTS CLASSIFICATION PROMPT:

Analyze this newspaper article about geopolitical risk. This article has been identified as discussing an adverse geopolitical event (Geopolitical risk). Geopolitical risk (GPR) encompasses adverse geopolitical events and associated risks, including:

- Wars (nuclear and conventional)
- Terrorist threats and acts
- Military buildups and tensions
- International crises and conflicts

TASK: Classify the nature of geopolitical risk discussed in this article. The three types of geopolitical risk nature are defined as follows:

1. threat: Risks, tensions, warnings, or potential escalations related to adverse geopolitical events
2. act: Actual adverse geopolitical events that have occurred or are actively occurring (wars, attacks, terrorist acts, invasions, coups, etc.)
3. both: Substantive discussion of both realized events AND associated threats/risks

Rules:

- Choose ONLY ONE: 'threat', 'act', or 'both'
- Be precise: only choose 'both' if the article substantially discusses BOTH temporal dimensions
- Brief mentions of past context while focusing on future risks = 'threat'
- Brief mentions of future risks while focusing on current/past events = 'act'

A.7.2 Oil GPR

The Oil GPR classification prompt first identifies whether an article talks about potential or realized oil supply disruptions. Relevant articles are then assigned regional labels based on where the oil supply disruption/threat is and which regions are affected. Multiple countries can be listed under the regional label.

OIL-GPR CLASSIFICATION PROMPT:

Analyze this newspaper article about geopolitical risk.

TASK 1: Determine if this article discusses events that could disrupt oil or energy supplies. Oil/Energy disruption includes:

- Military conflicts in oil-producing regions
- Sanctions on oil-producing countries
- Attacks on oil infrastructure, pipelines, refineries, or shipping routes
- Political instability in major oil producers
- Blockades of oil shipping lanes (Strait of Hormuz, Suez Canal, etc.)
- Natural disasters affecting oil production
- Oil embargoes or export restrictions

- Terrorist attacks on energy infrastructure

Answer "yes" if the article discusses events that could affect oil/energy production, transportation, or supply. Answer "no" if the article discusses geopolitical events unlikely to affect oil/energy markets.

TASK 2: If "yes", identify which oil/energy producing region(s) are affected. Select one or more from this list ONLY: Middle_East, Russia, USA, Venezuela, North_Africa, West_Africa, Central_Asia, North_Sea, Canada, Mexico, Latin_America, Southeast_Asia, China, Other

REGIONAL MAPPING RULES:

- Use 'Middle_East' for: Saudi Arabia, Iran, Iraq, Kuwait, UAE, Qatar, Bahrain, Oman, Yemen, Israel, Syria, Lebanon
- Use 'Russia' for: Russia, Soviet Union, USSR, Russian Federation
- Use 'USA' for: United States, America, USA, U.S. (as oil producer, not just consumer)
- Use 'Venezuela' for: Venezuela
- Use 'North_Africa' for: Libya, Algeria, Egypt, Tunisia
- Use 'West_Africa' for: Nigeria, Angola, Equatorial Guinea, Gabon, Congo
- Use 'Central_Asia' for: Kazakhstan, Azerbaijan, Turkmenistan, Uzbekistan
- Use 'North_Sea' for: Norway, UK (when referring to North Sea oil production)
- Use 'Canada' for: Canada (oil sands, Alberta, etc.)
- Use 'Mexico' for: Mexico
- Use 'Latin_America' for: Other Latin American oil producers (Brazil, Colombia, Ecuador, Argentina)
- Use 'Southeast_Asia' for: Indonesia, Malaysia, Brunei, Vietnam
- Use 'China' for: China (as oil producer)
- Use 'Other' for: Any other oil/energy producing region not listed above

IMPORTANT:

- Focus on regions WHERE the disruption is occurring, not where it's being discussed
- Include ALL regions mentioned that could experience oil/energy disruption
- If article mentions global impact or multiple regions, list all relevant ones
- If uncertain whether it affects oil/energy, err on the side of 'yes' if the region is a major producer

A.7.3 Country GPR (and Event)

The country classifier prompt identifies the main countries involved in a geopolitical incident as portrayed in a given article. Relevant countries are assigned one of three actor labels indicating whether they are an initiator, a respondent, or a spillover party. The prompt also assigns an event-type label for the article's central geopolitical conflict.

COUNTRY GPR:

Analyze this newspaper article about a geopolitical risk event.

TASK: Identify the key country-level actors in this geopolitical event and assign each a role. Roles are defined as follows:

- initiator: the country that launched or triggered the hostile action (military attack, invasion, sanctions, blockade, etc.)
- respondent: the country that is the direct object of the hostile action
- spillover: a country not directly involved but significantly affected (energy shock, refugee flows, trade disruption, etc.)

Rules:

- List up to 5 actors total; include only countries with a meaningful role in THIS article
- Use standard country names (e.g. "United States", "Russia", "Saudi Arabia", "Israel")
- If a supra-national body (NATO, UN, EU) is the dominant actor, use its name
- Assign exactly one role per actor
- When initiation is genuinely contested or unclear (civil wars, mutual escalations, disputed territories), assign both main parties the role of initiator
- If the article is too vague to identify actors, return an empty actors list

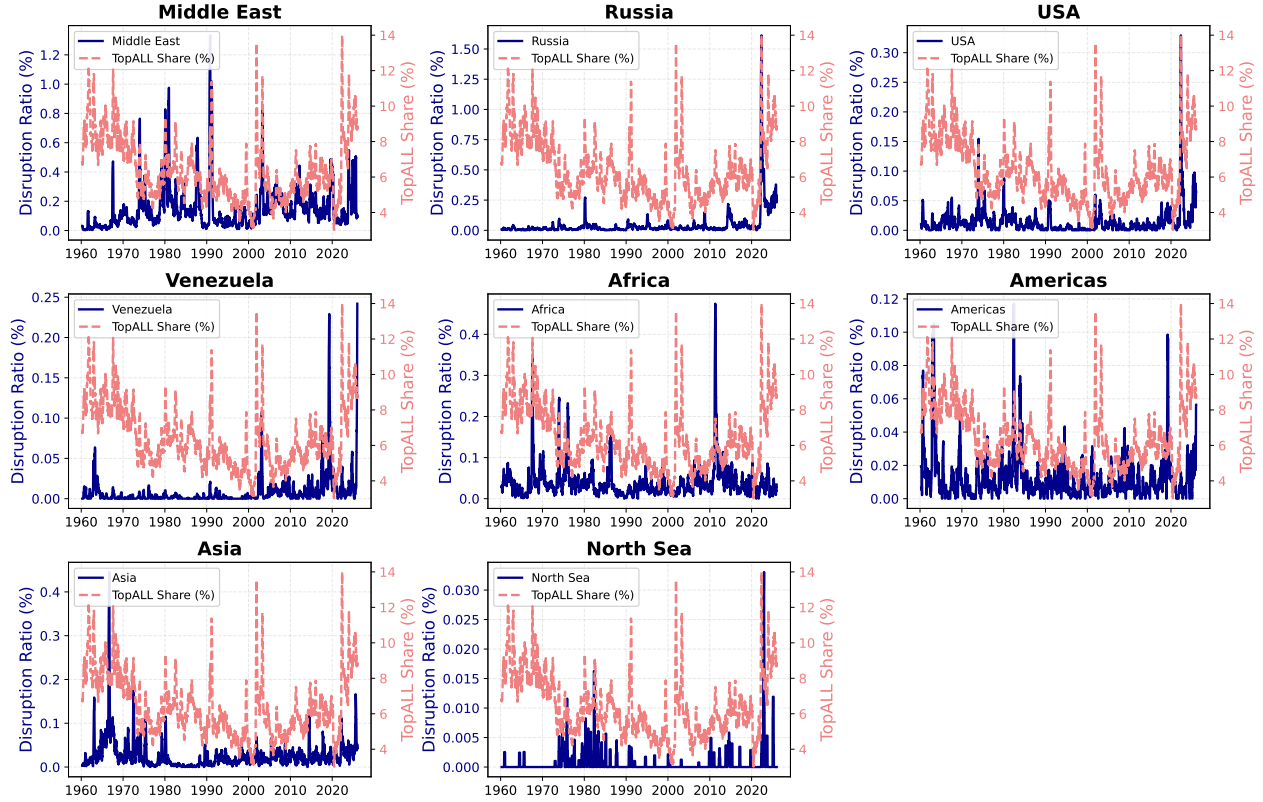
Respond in JSON format only:

```
{"actors": [
  {"country": "<name>", "role": "<role>"},
  ...
],
"event_type": "<one of: military_conflict, sanctions, terrorism,
diplomatic_tension, coup, civil_war, nuclear_threat, other>"}
```

A.8 GPR-Driven Oil Disruptions by Region

This appendix shows a decomposition of the Oil GPR index of Section 5.2 by geographic region. For each region, we apply the same two-stage LLM classification used for the aggregate index but retain only articles in which the oil or energy disruption occurs in, or affects, that region; each series is the sentiment-weighted count of such articles divided by total newspaper volume. Figure A.3 reports the eight regional panels.

Figure A.3: GPR-DRIVEN OIL DISRUPTIONS BY REGION



Notes: Each panel shows the regional Oil GPR index (90-day moving average) for a specific geographic region. The index is computed as the ratio of sentiment-weighted articles about oil disruptions in that region to total newspaper articles. The light red dashed line in each panel shows the overall AI-GPR share for reference. Regions are: Middle East, Russia, USA, Venezuela, Africa (North + West), Americas (Canada, Mexico, Latin America), Asia (Central, Southeast, China), and North Sea. Source: authors' calculations using the two-stage LLM classification described in Section 5.2. Last observation: December 2025.

A.9 Data Details for the Oil Analysis

This appendix collects the data sources behind Figure 5 and the estimation choices behind the proxy VAR of Section 6.2. In the figure, the oil price is the WTI spot price, FRED series WTISPLC, topped up for the current partial month with the daily series DCOILWTICO, and deflated by the consumer price index, FRED series CPIAUCSL, to the dollars of the last month plotted. The system is estimated in levels with a constant and 12 lags, on monthly data from January 1985 to May 2026. The oil price is the U.S. crude oil first purchase price published by the Energy Information Administration, in nominal dollars per barrel and entered in logs. Global oil production is world crude oil including lease condensate, in thousand barrels per day, from the same source; it is the series that determines how far the sample can be extended. Consumer prices enter in logs and the ten-year Treasury yield in levels, both from FRED. The daily oil price used to build the instrument is a different series: it is the WTI spot price at Cushing, which is quoted daily and is available from 1986 onward.

Monthly GDP is the Brave-Butters-Kelley index, which is published as an annualized monthly log difference. We divide it by 1,200 to convert it to a monthly log change and cumulate it, so that it enters the VAR as a log level comparable to the other variables. The Oil GPR threats index enters the VAR as a monthly average of the daily series, standardized over the full history of the index rather than over the estimation sample.

The consumer price index has one missing observation, October 2025, which the Bureau of Labor Statistics did not publish because of the federal government shutdown. We interpolate it linearly between September and November 2025. No other variable in the system is interpolated, and the ragged end of each series is left as is.

Flagged days are frequent enough that the instrument is not sparse: the Oil GPR threats index exceeds one standard deviation on about nine percent of days, which leaves roughly two thirds of the months in the sample with a non-zero instrument. Confidence bands come from the moving block bootstrap of [Jentsch and Lunsford \(2022\)](#) with 6-month blocks and 500 draws. Longer blocks leave too few blocks per draw for an instrument whose identifying variation is concentrated in a small number of days, and the resulting bands are no longer centered on the point estimate.